Deploying Metabase on AWS

by Mohamad Ikhsan Nurulloh

Disclaimer

- **No Endorsement:** This guide is for educational purposes only and does not imply endorsement by Metabase or AWS.
- **Testing Only:** Intended for testing and development, not production environments.
- **Refer to Official Docs:** Always consult official Metabase and AWS documentation for production use.
- **Proceed at Your Own Risk:** Use this guide at your own risk and be aware of potential deployment implications.

What is Metabase?

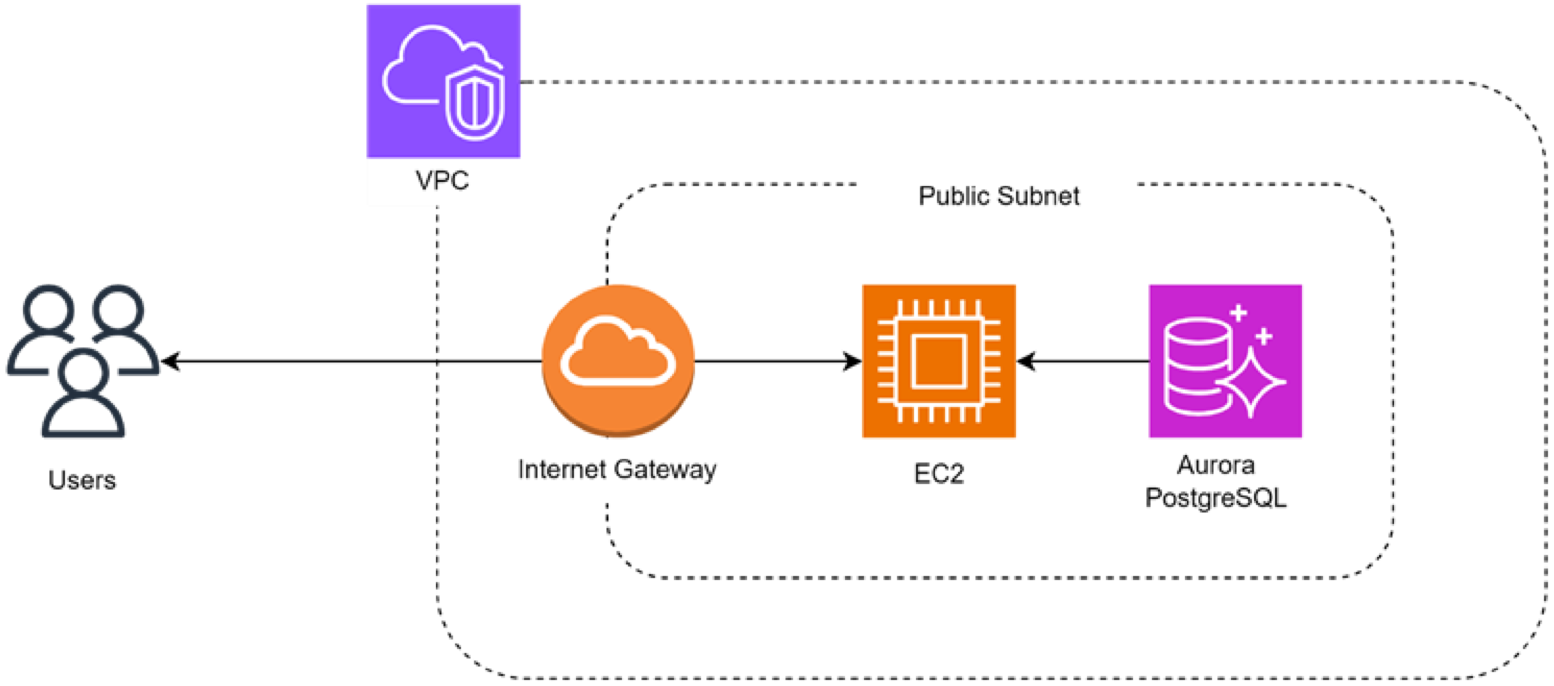
Metabase is an open-source BI tool for visualizing data through charts and dashboards. It offers a Query Builder, customizable dashboards, and alerts. Ideal for business reporting, quick analysis, and data exploration, Metabase simplifies data-driven decision-making.

Why Metabase

- **User-Friendly:** Metabase’s intuitive interface lets users create and share dashboards without extensive technical knowledge.
- **Open Source:** Free and customizable, Metabase benefits from a large, active community.
- **Powerful Analytics:** Offers robust analytics, complex queries, and diverse data visualizations.
- **Seamless Integration:** Integrates smoothly with various databases and data sources.
- **Collaboration:** Promotes teamwork by enabling easy sharing of dashboards and reports.

Metabase Architecture

Metabase Architecture



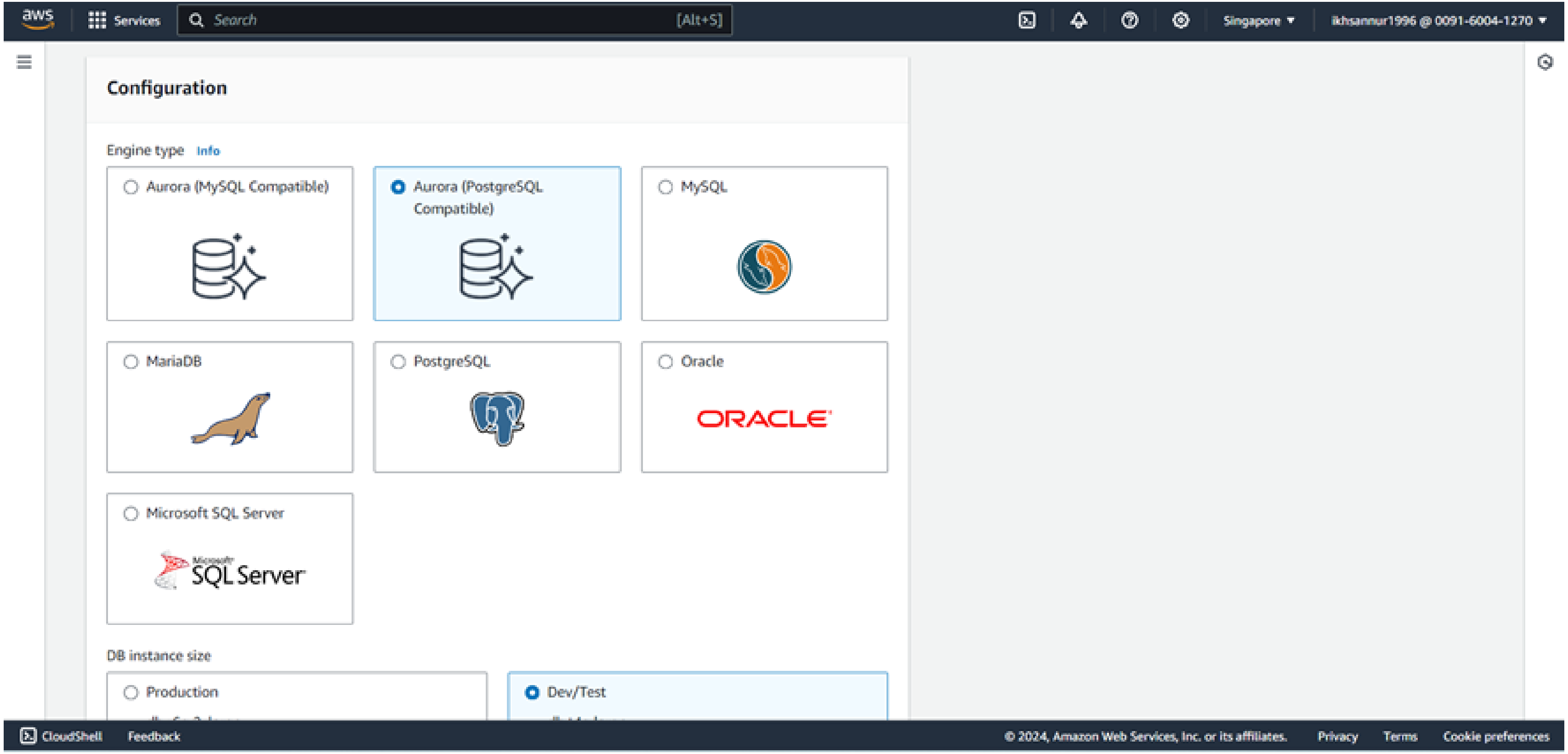
Overall Architecture: Users access the Metabase application hosted on an EC2 instance in a public subnet. The EC2 instance communicates with the Aurora PostgreSQL database within the VPC to retrieve and analyze data. The internet gateway allows the EC2 instance to be accessible from the internet, enabling users to interact with Metabase from anywhere.

Details Metabase Architecture

- **Users:** End-users (data analysts, business users) who interact with Metabase to query data, generate reports, and visualize insights.
- **VPC (Virtual Private Cloud):** A secure and scalable AWS environment for deploying resources like EC2 and RDS, ensuring network isolation and secure communication.
- **Internet Gateway:** Enables EC2 instances in the VPC to connect to the internet, allowing users to access Metabase from anywhere.
- **EC2 (Elastic Compute Cloud):** Hosts the Metabase application, processing user requests, querying databases, and serving the interface.
- **Public Subnet:** A subnet that allows the EC2 instance running Metabase to be accessible from the internet.
- **Aurora PostgreSQL:** A managed PostgreSQL database engine that stores and supports data queries and analysis in Metabase.

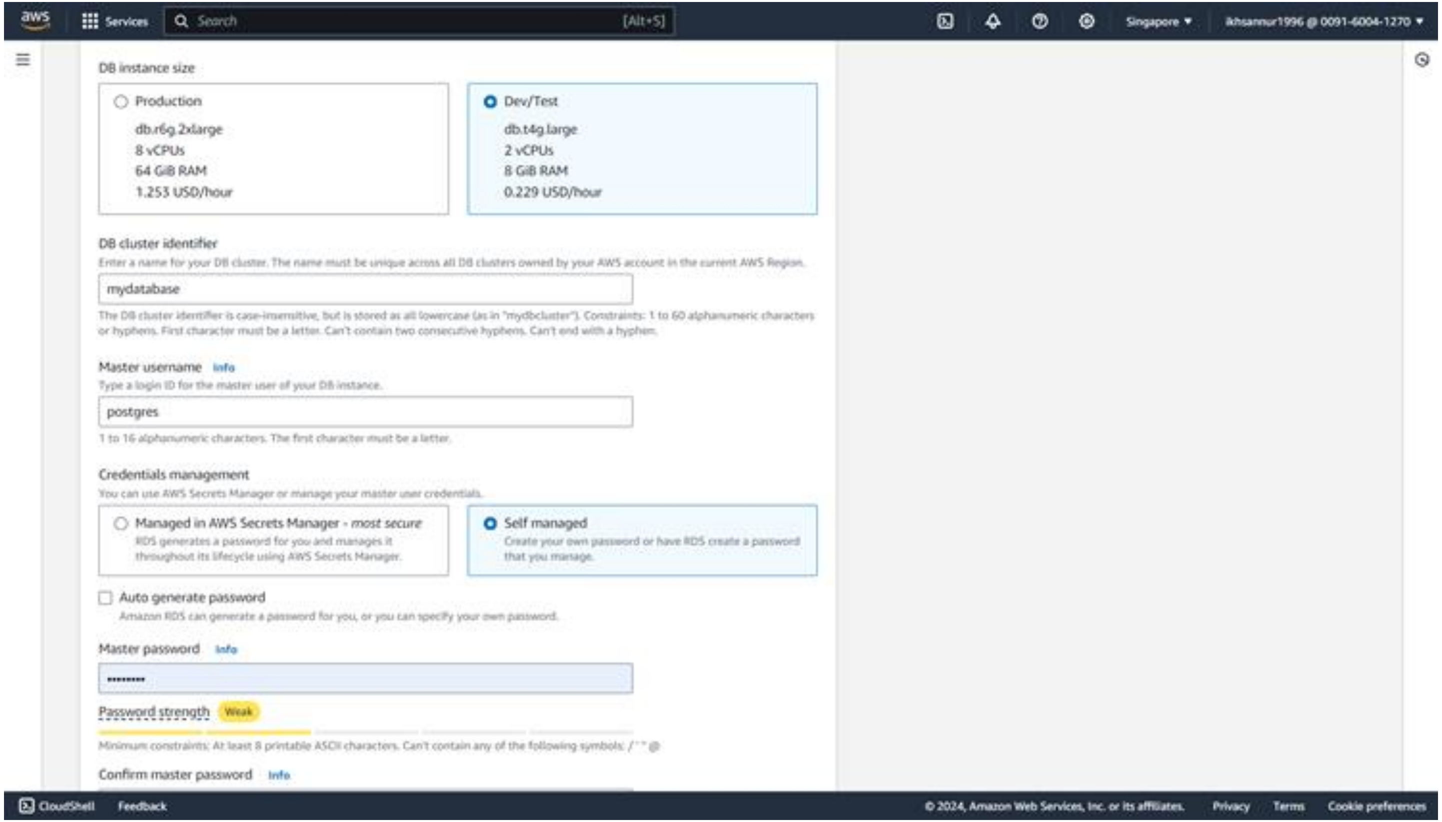
Setting Up RDS Aurora (PostgreSQL Compatible)

Creating RDS - Aurora (PostgreSQL Compatible)



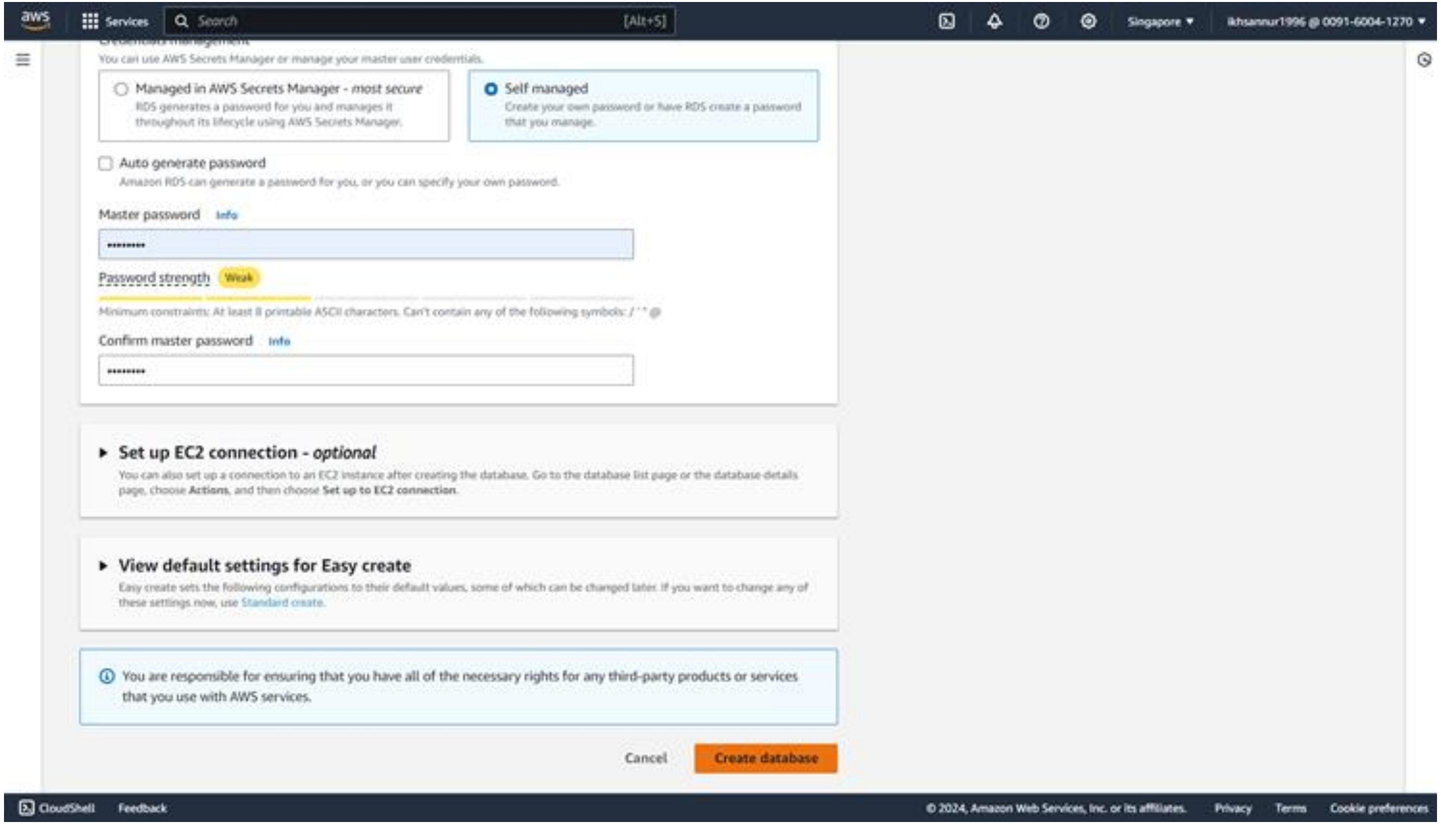
Detailed steps for creating an Amazon RDS instance using Aurora PostgreSQL, including selecting the engine type and configuring the database instance.

Select DB Instance Size



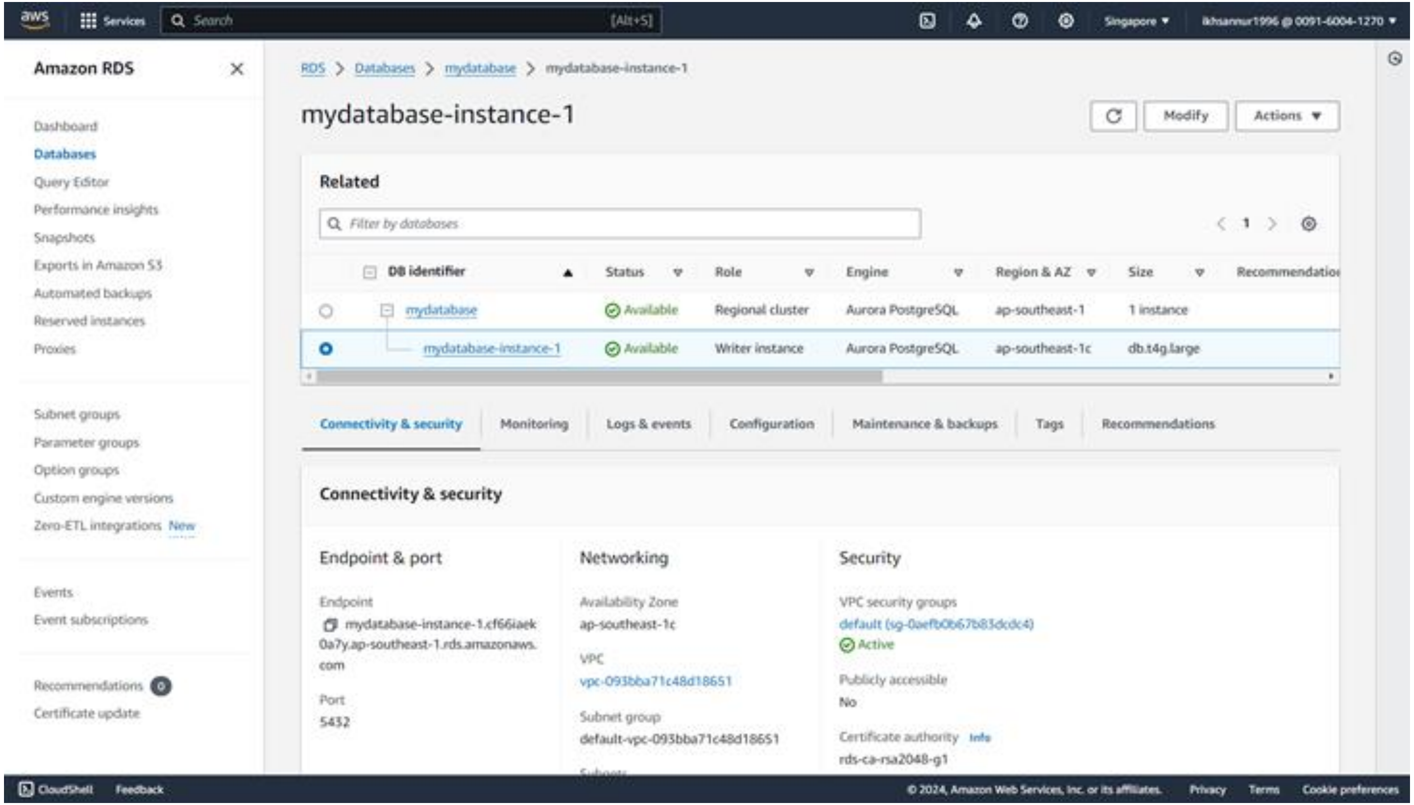
This page covers the configuration options for the DB instance, including choosing the instance size (e.g., production or dev/test), specifying the DB cluster identifier, and setting the master username.

Credentials Management



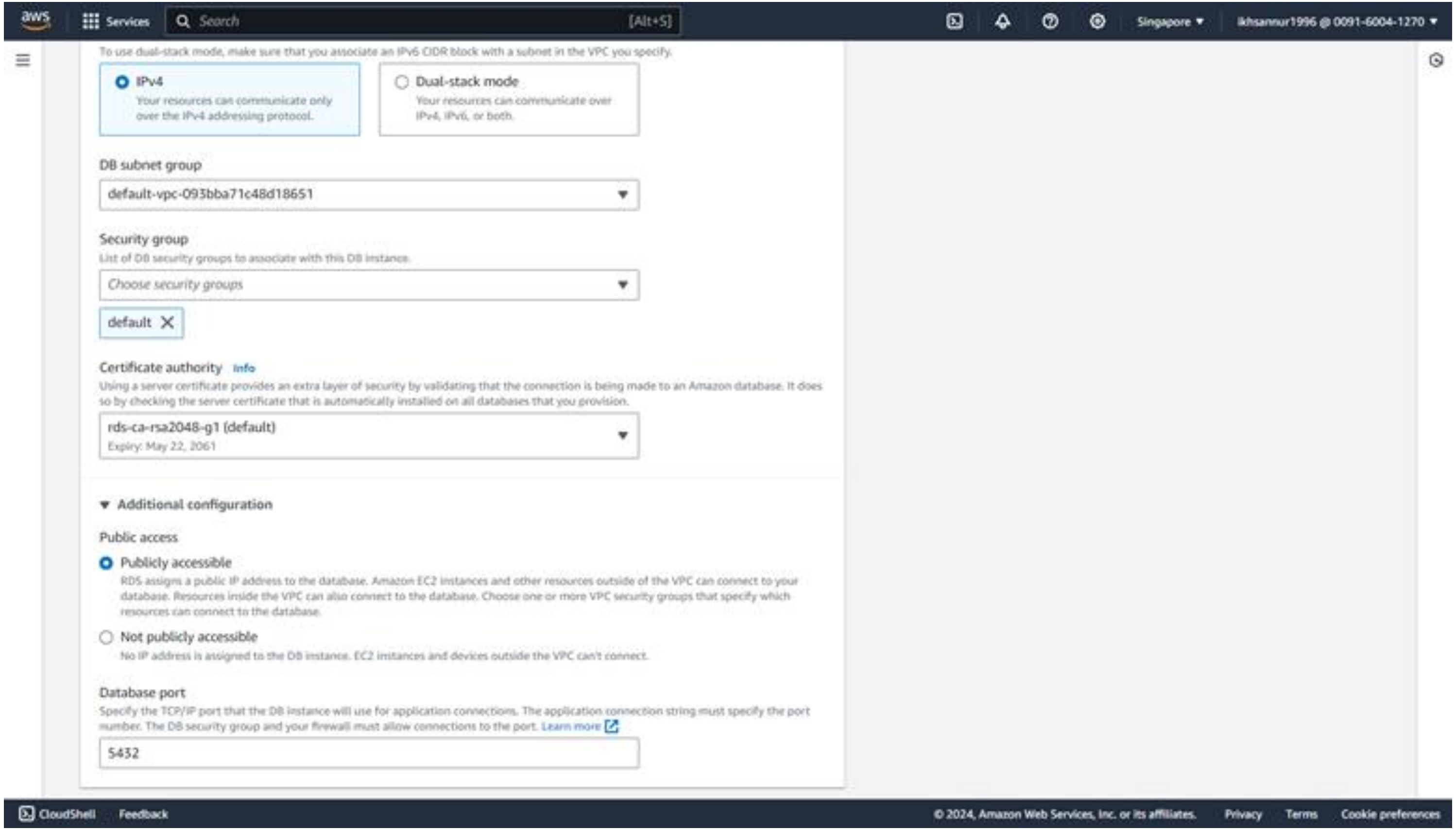
Options for managing master user credentials, including using AWS Secrets Manager for secure password management or self-managing the credentials.

RDS Aurora PostgreSQL Success



Confirmation of the successful deployment of the RDS Aurora PostgreSQL instance, with details about the database instance, such as endpoint, port, and availability zone.

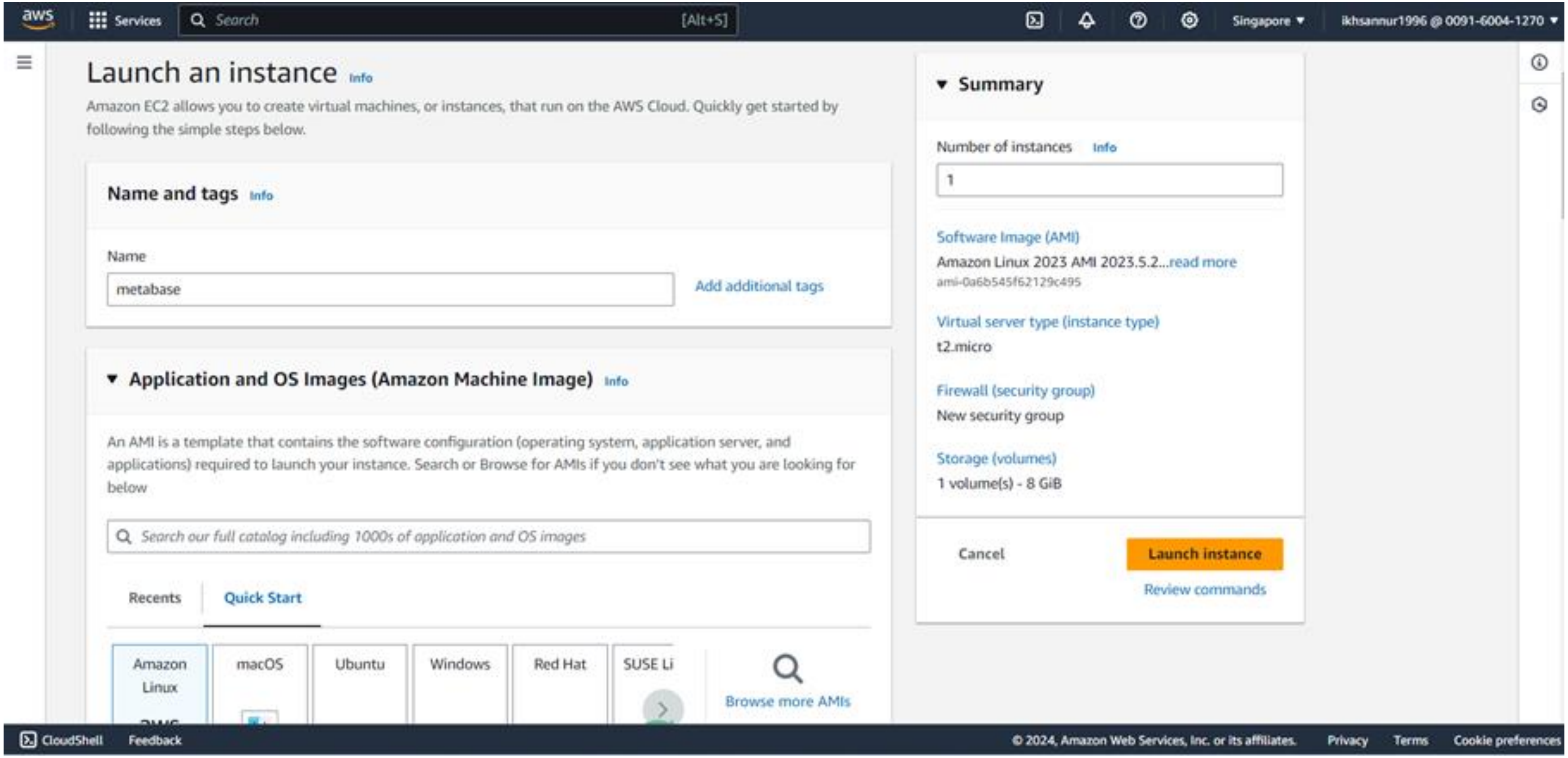
Enable Public Access



Enabling public access to the database and configuring security groups.

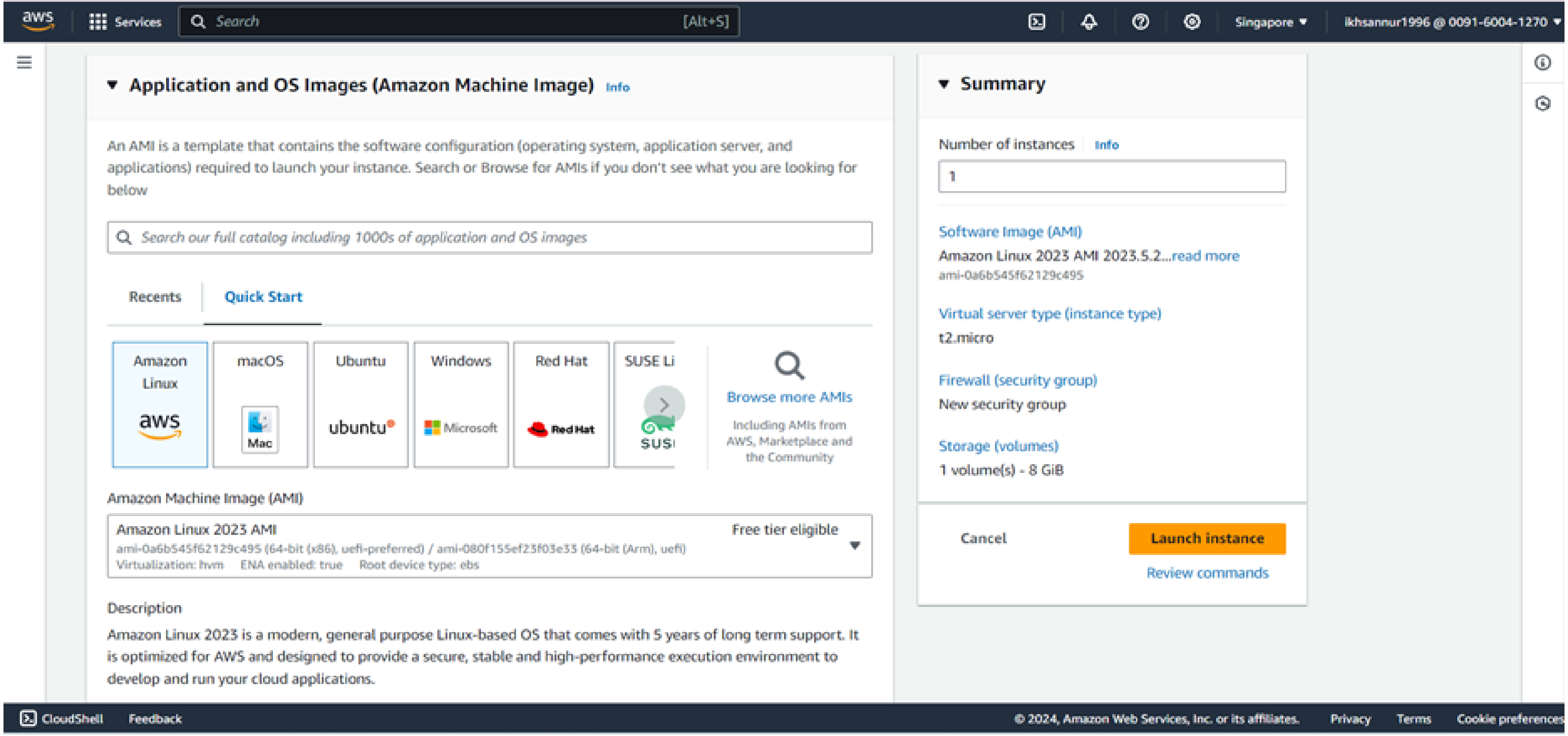
Setting Up Amazon EC2

Launch EC2 Instance



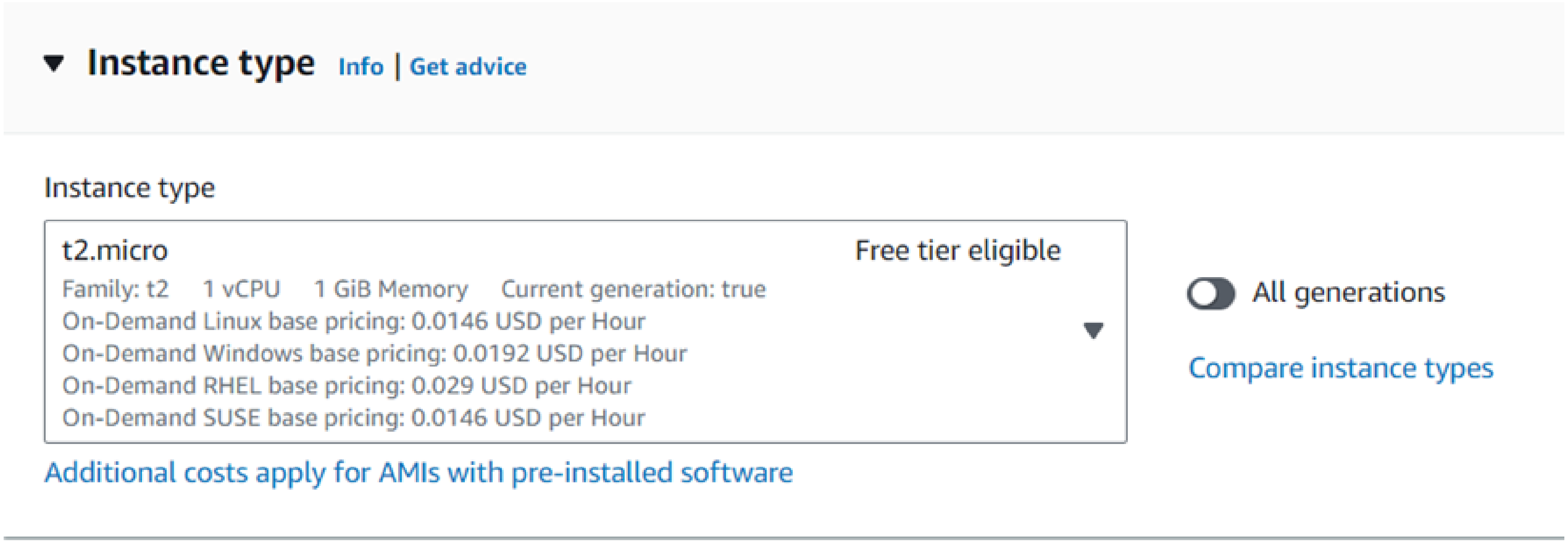
Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

Choose Amazon Machine Image



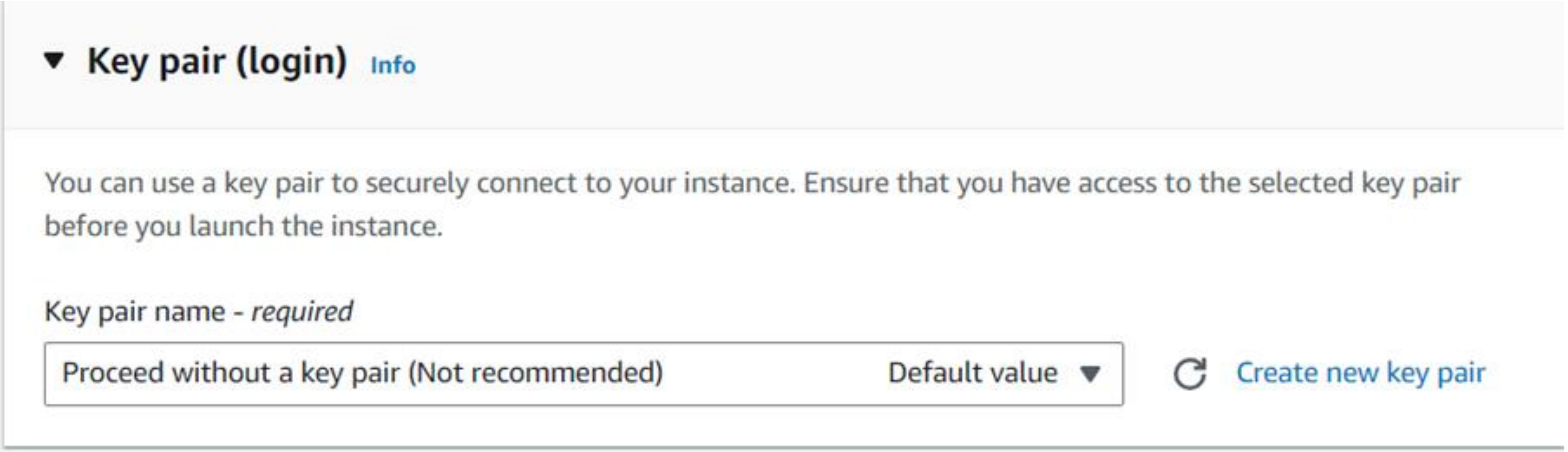
Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

Select Instance Type



Selecting the Amazon Machine Image (AMI), instance type, and configuring security group settings.

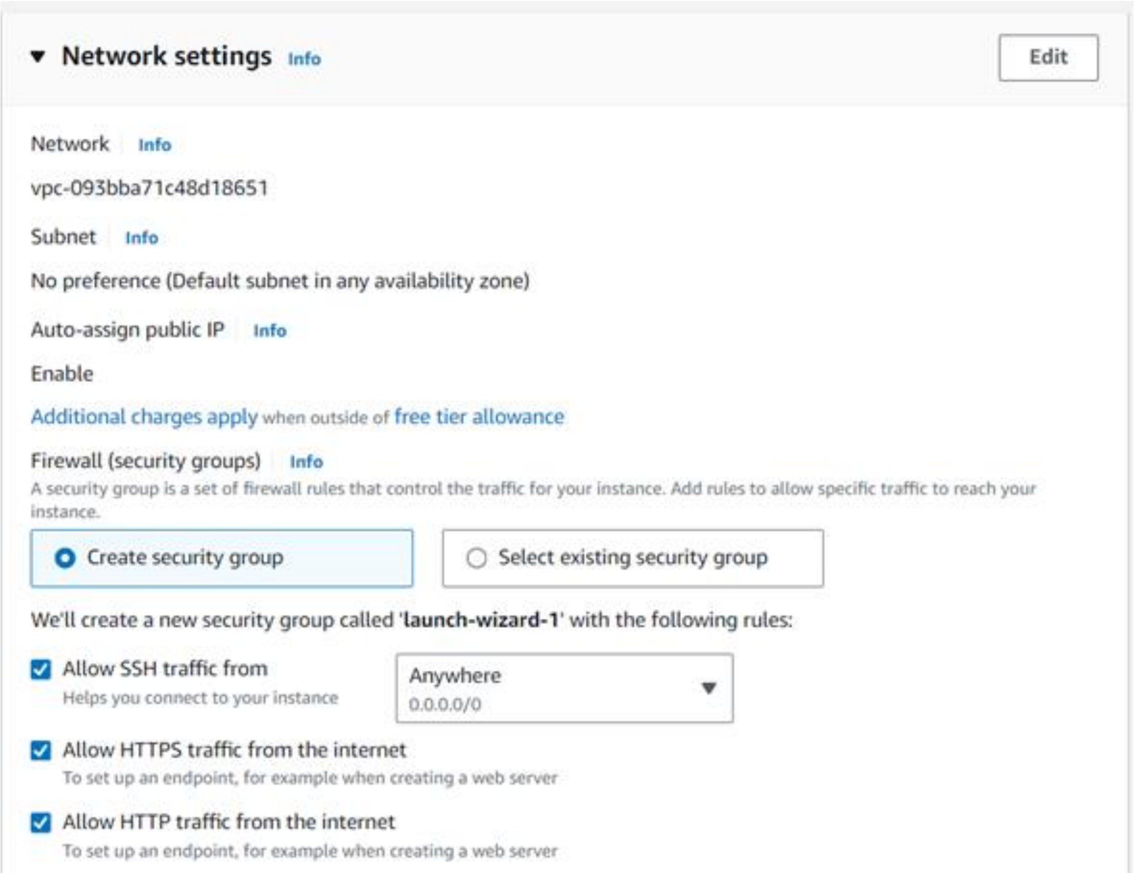
Key Pair Configuration



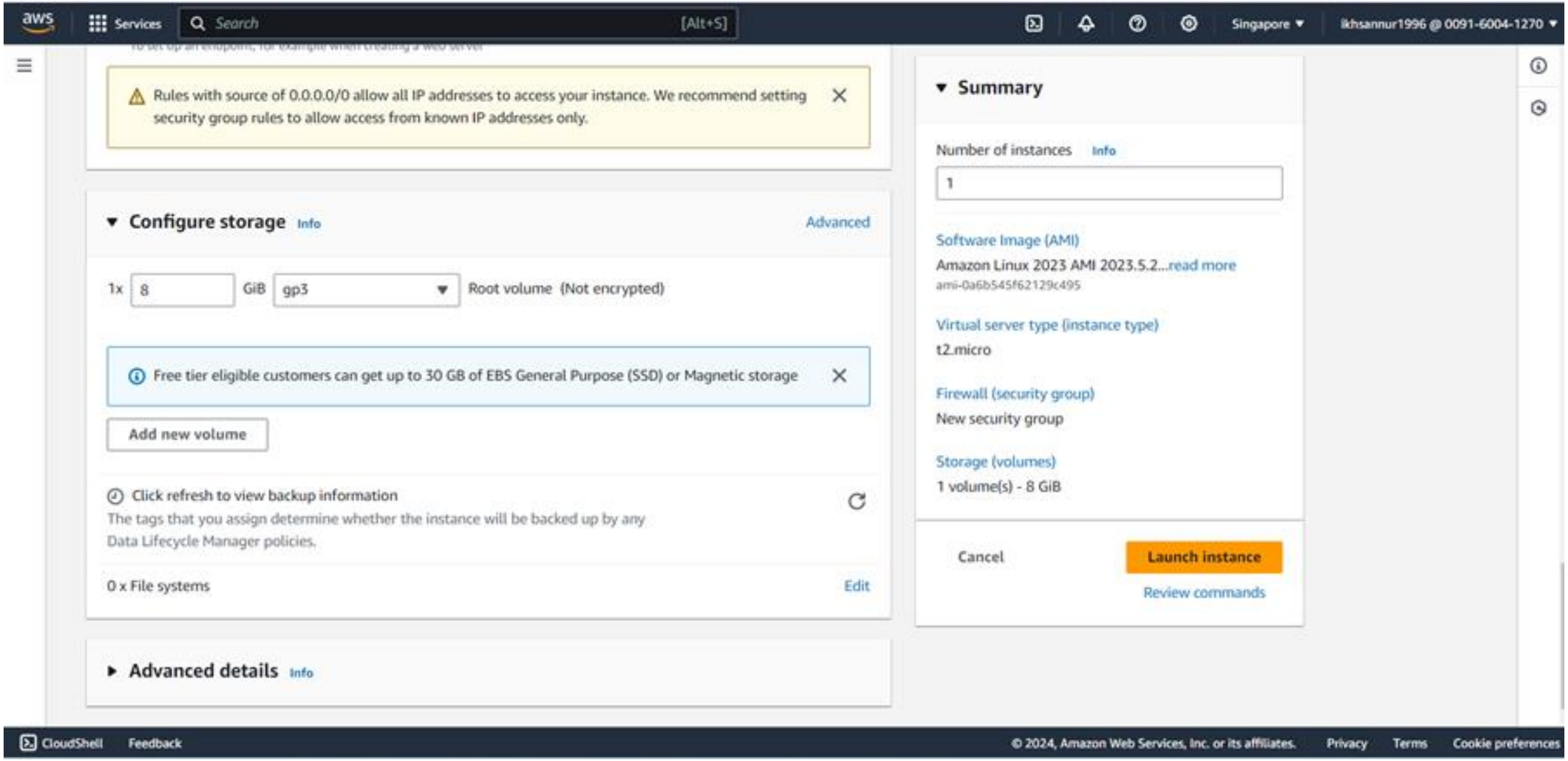
Options for managing AWS Key Pair for secure login management.

EC2 Network Configuration

Configuration of security group rules to allow access to the Metabase port, including setting up rules for HTTP, HTTPS, and SSH traffic.

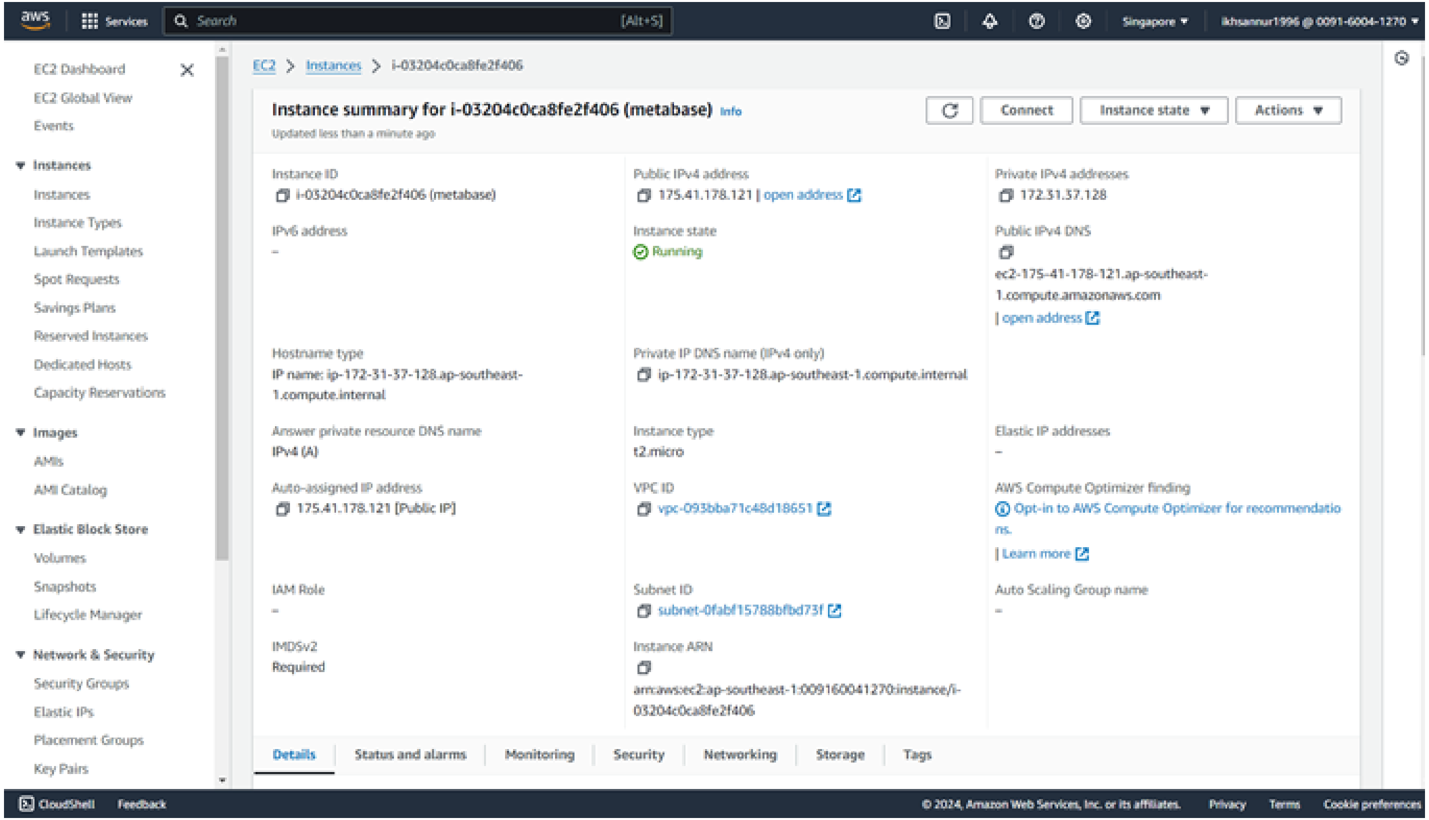


Storage Configuration



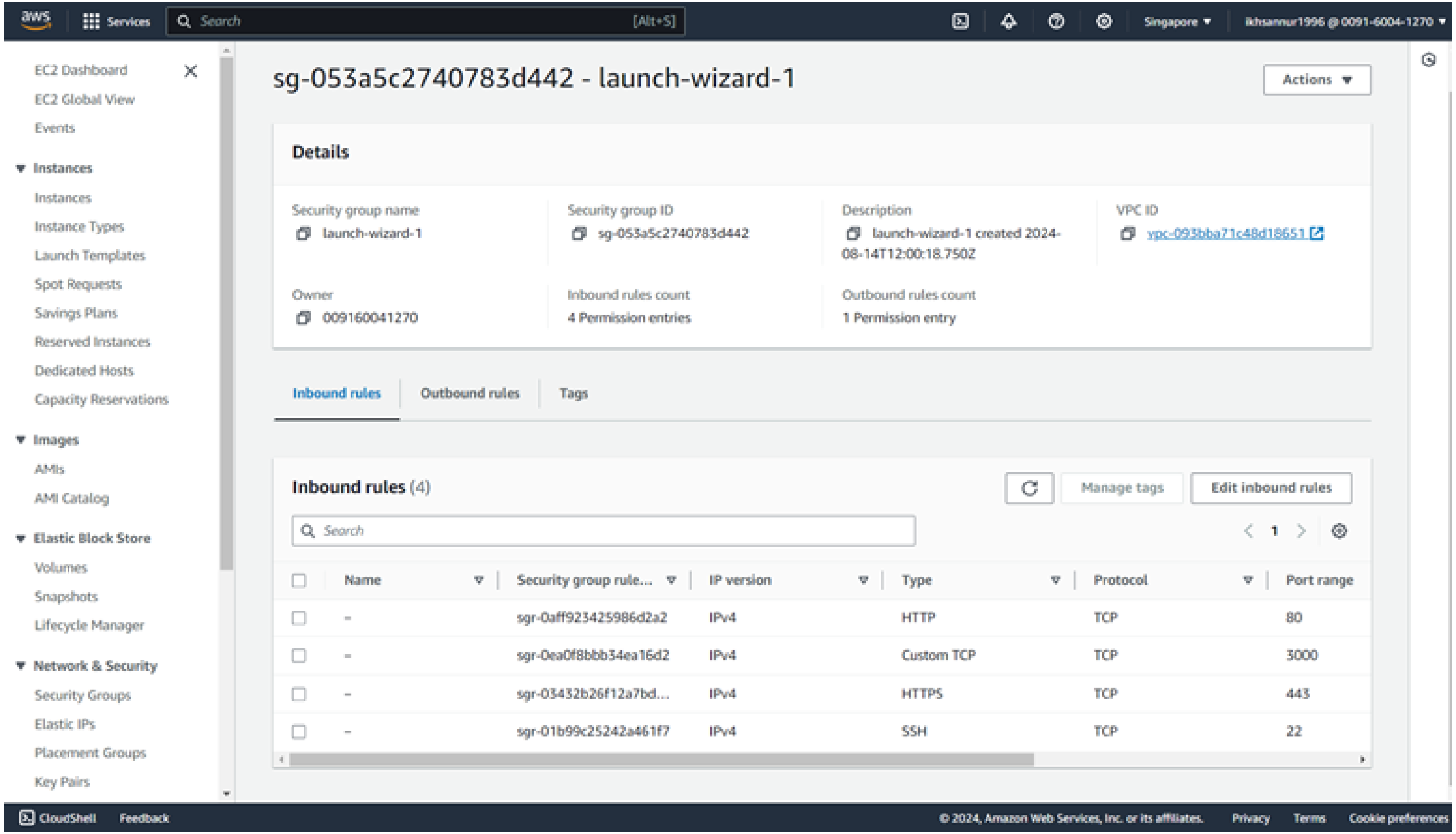
Options for Configuring Storage Size.

EC2 Instance Success Running



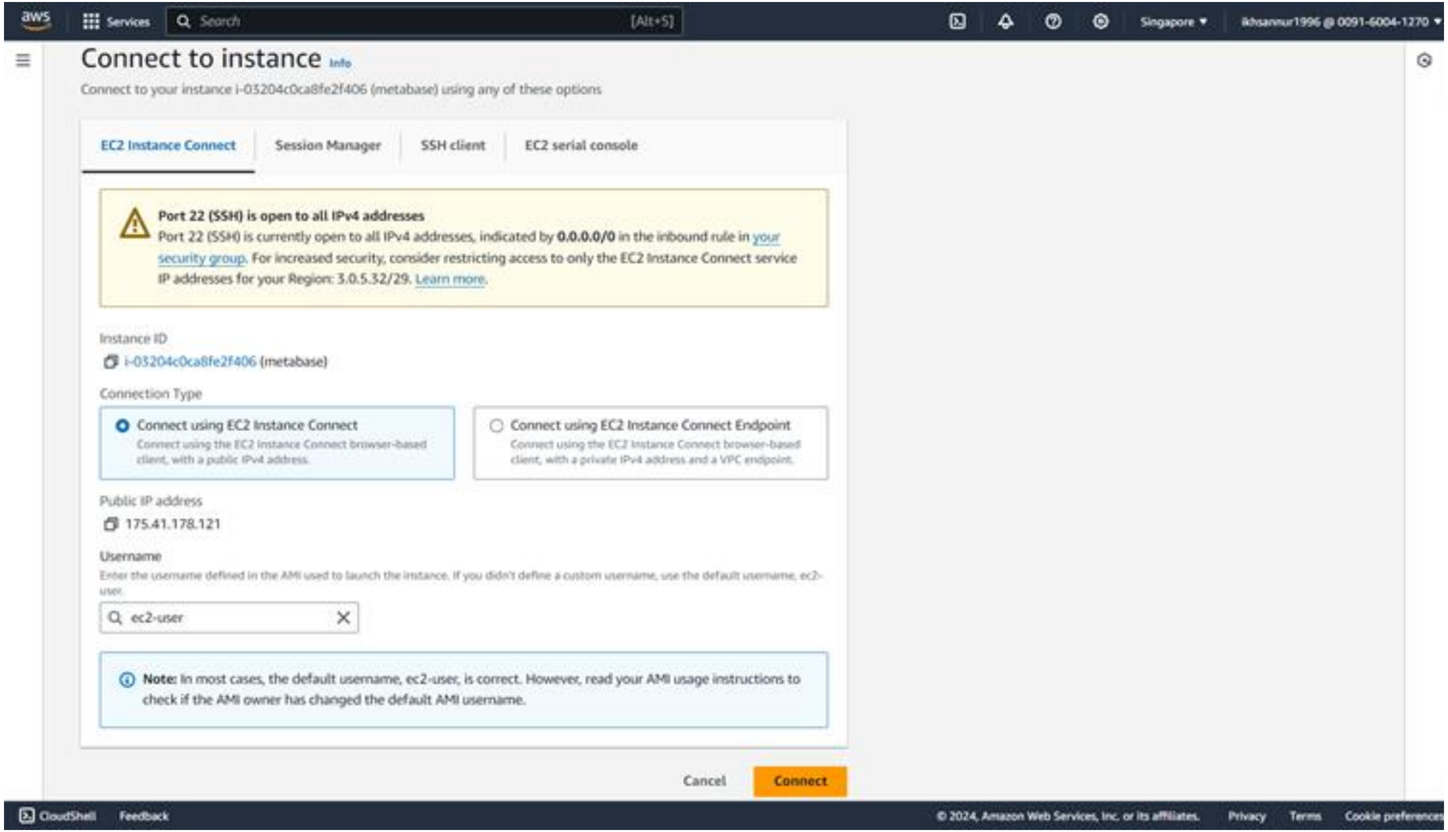
Confirmation of the successful launch of the EC2 instance, with details about the instance, such as instance ID, public and private IP addresses, and instance state.

Custom Inbound Rules for Metabase Port



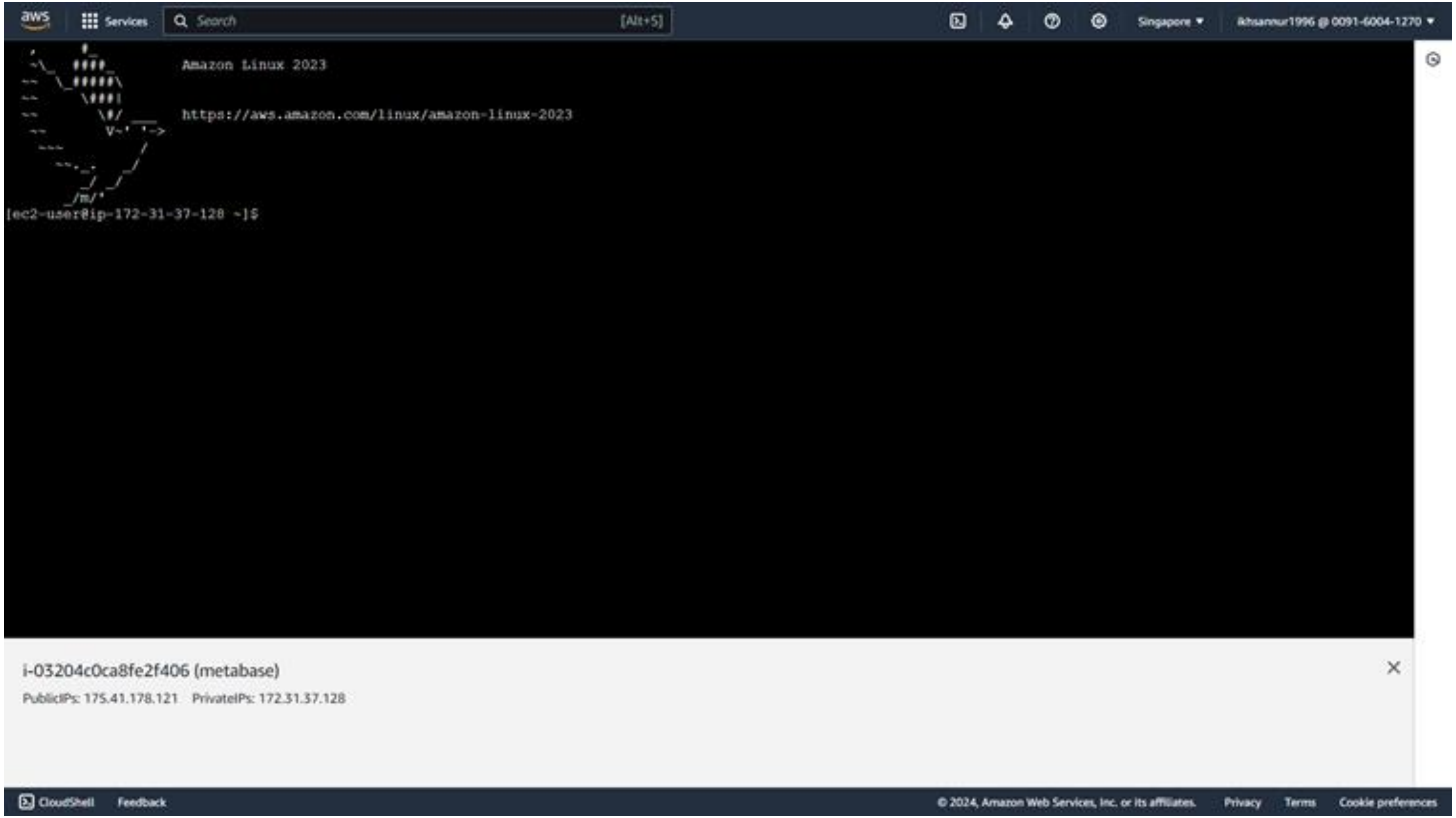
Configuration of security group rules to allow access to the Metabase port, including setting up rules for HTTP, HTTPS, and SSH traffic.

Connect to Instance



Instructions for connecting to the EC2 instance using various methods, such as EC2 Instance Connect, Session Manager, SSH client, and EC2 serial console.

SSH Opened



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Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-37-128 ~]$
i-03204c0ca8fe2f406 (metabase)
PublicIPs: 175.41.178.121 PrivateIPs: 172.31.37.128
CloudShell
Feedback
© 2024, Amazon Web Services, Inc. or its affiliates.
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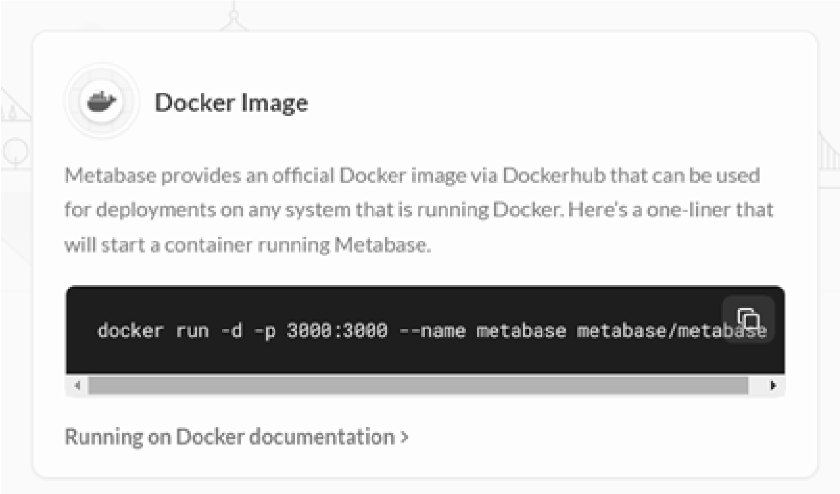
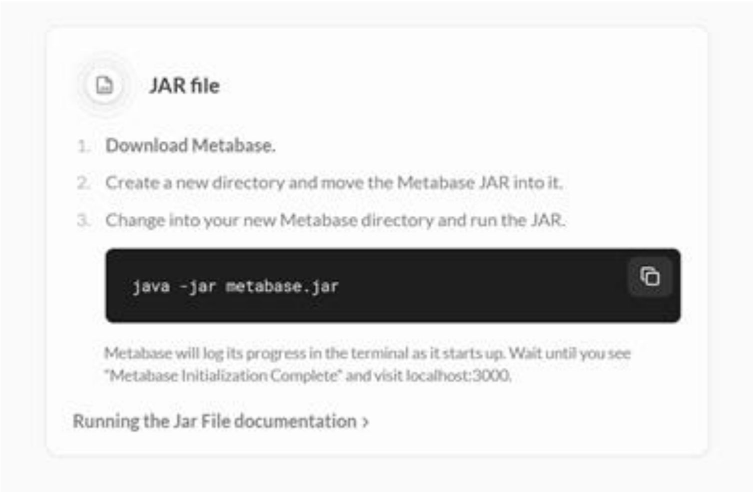
Setting Up Metabase on Amazon EC2

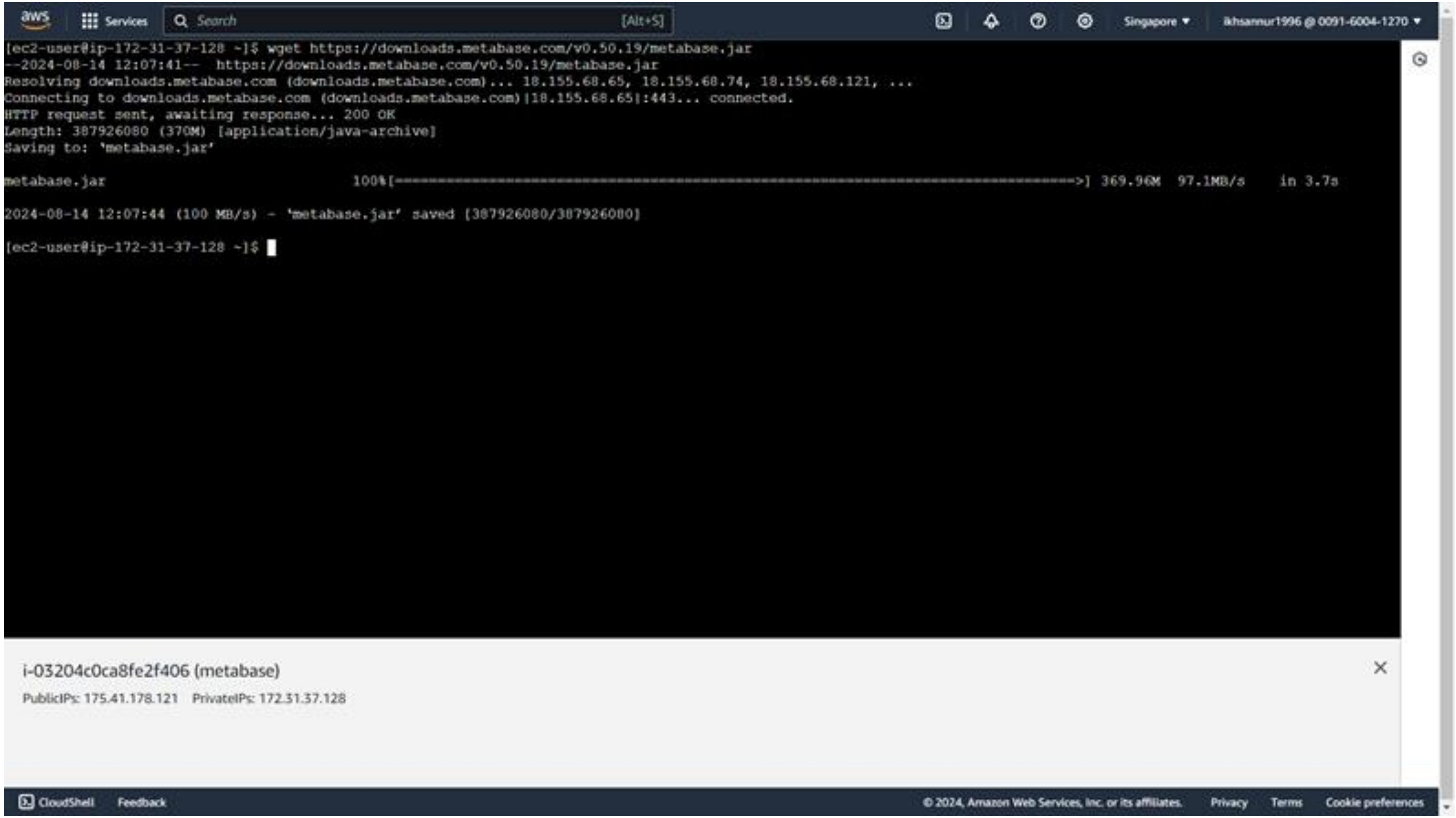




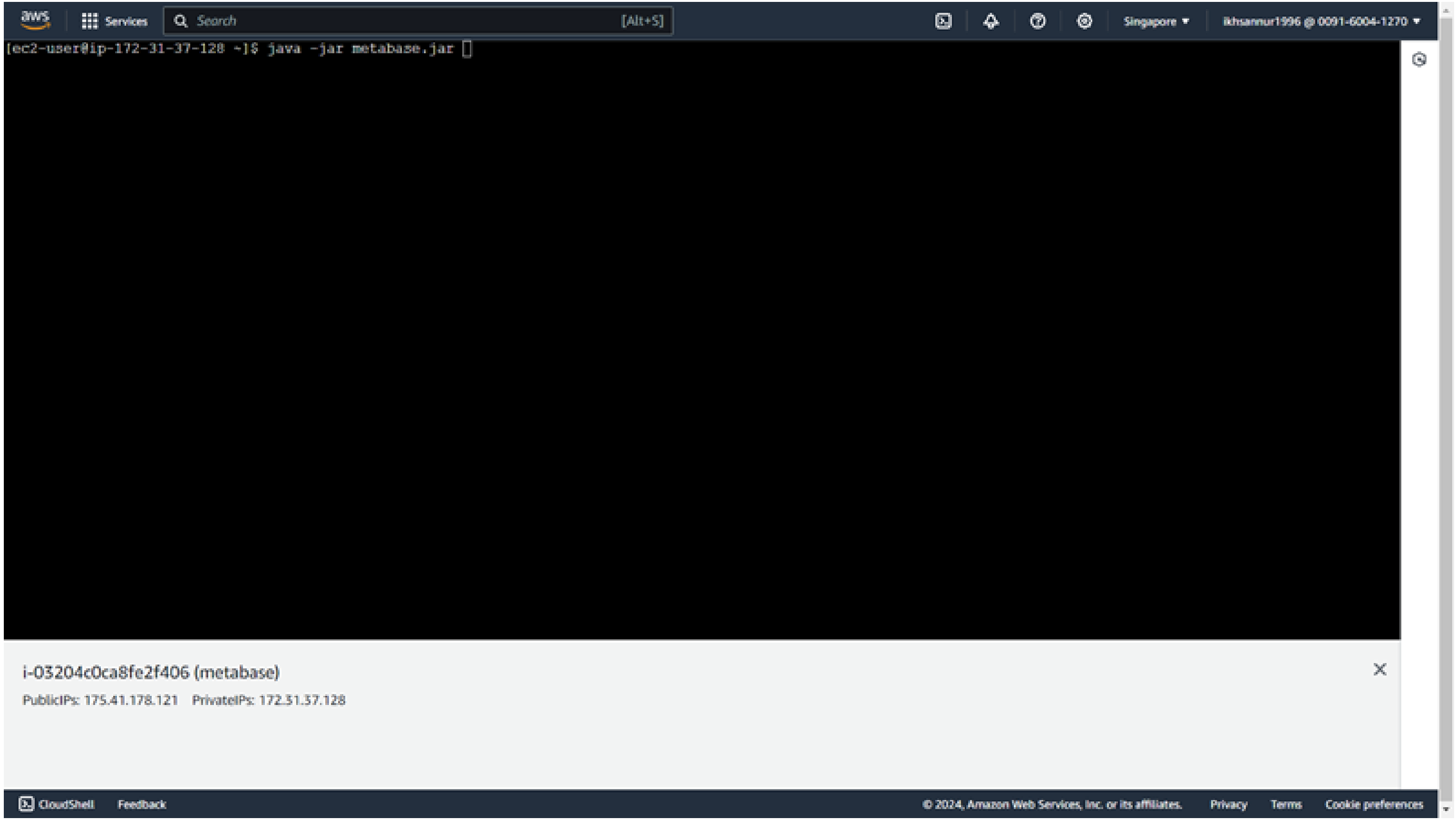
Steps to install Metabase on the EC2 instance, including downloading the Metabase application and setting up the necessary configurations. If Java isn't installed, you'll need to install Java before you can run Metabase.

https://www.metabase.com/start/oss/

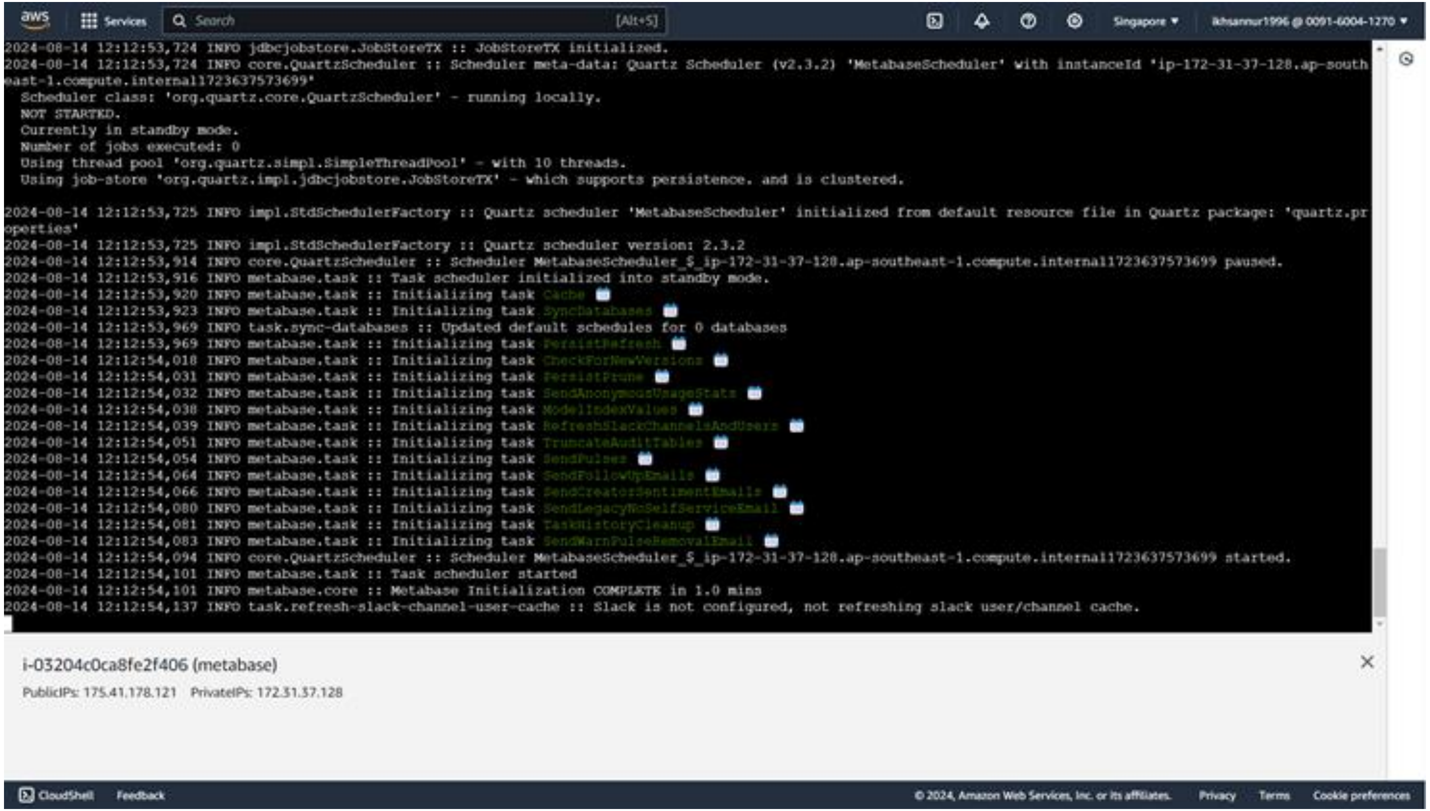
Downloading Metabase



Run The Metabase



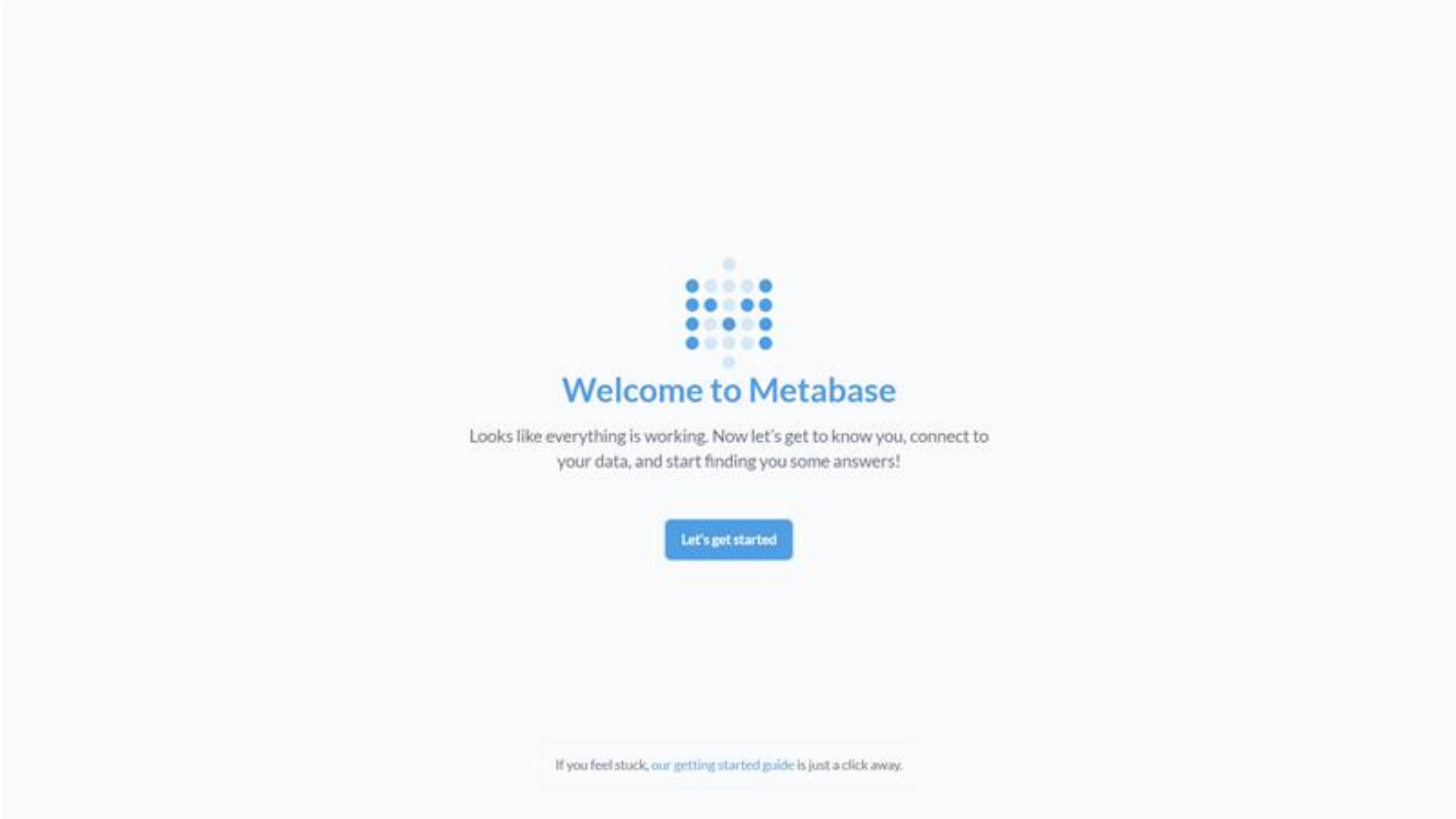
Metabase Success Running



Logs and details of the Metabase initialization process, including the setup of the Quartz scheduler and various Metabase tasks. Metabase will run in port 3000.

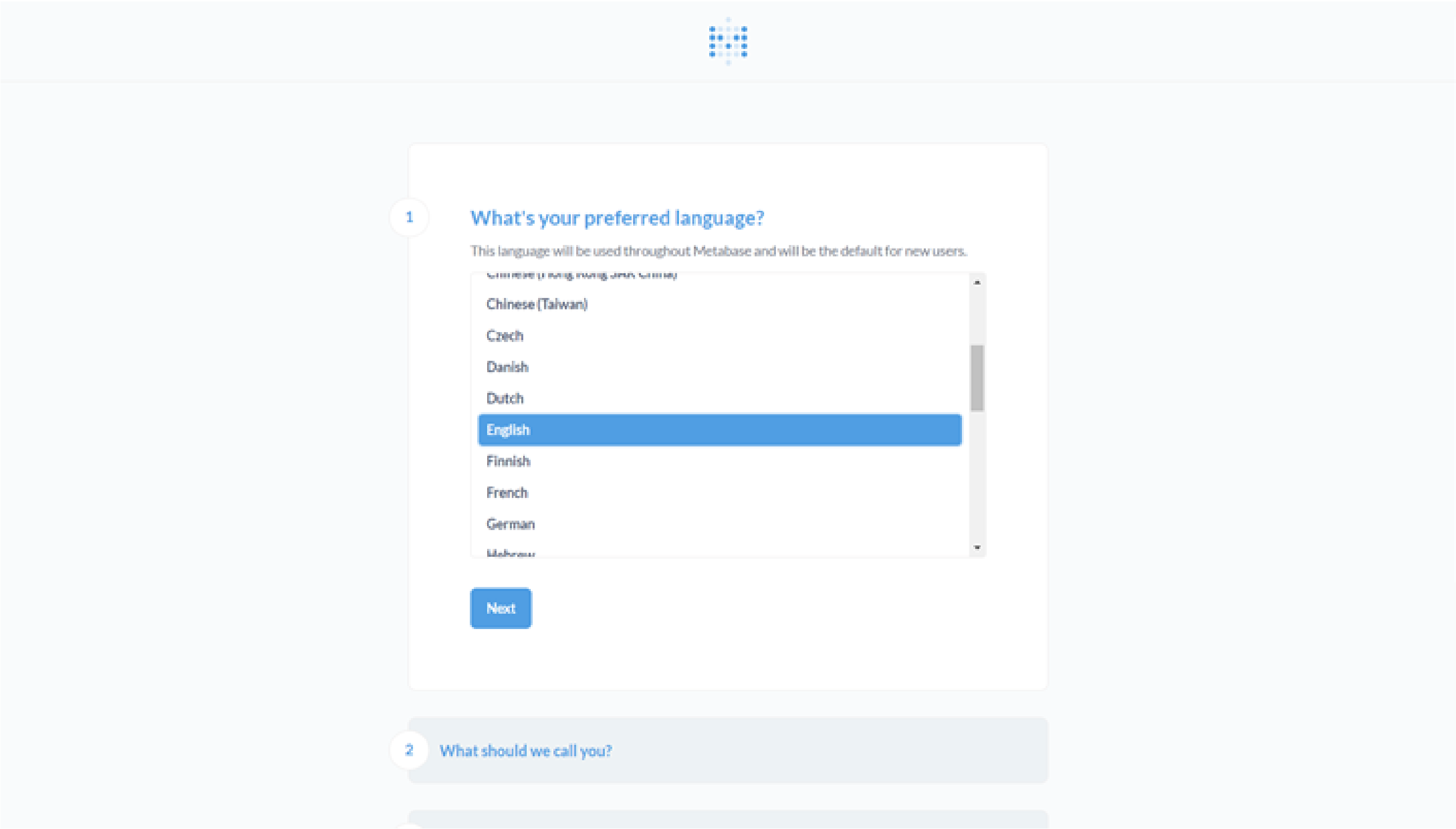
Metabase Initialisation and Configuration

Welcome to Metabase



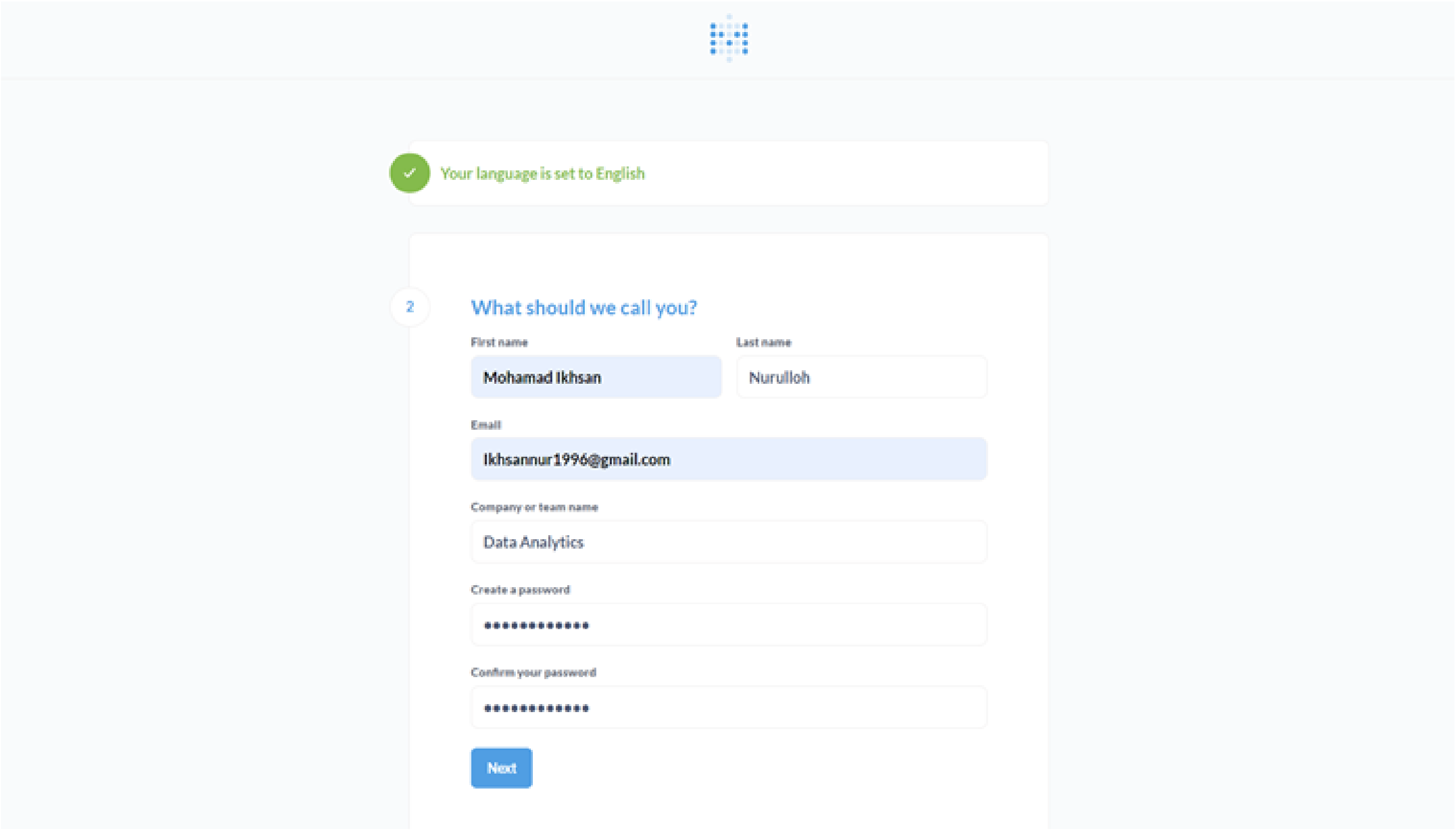
Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

Language Configuration



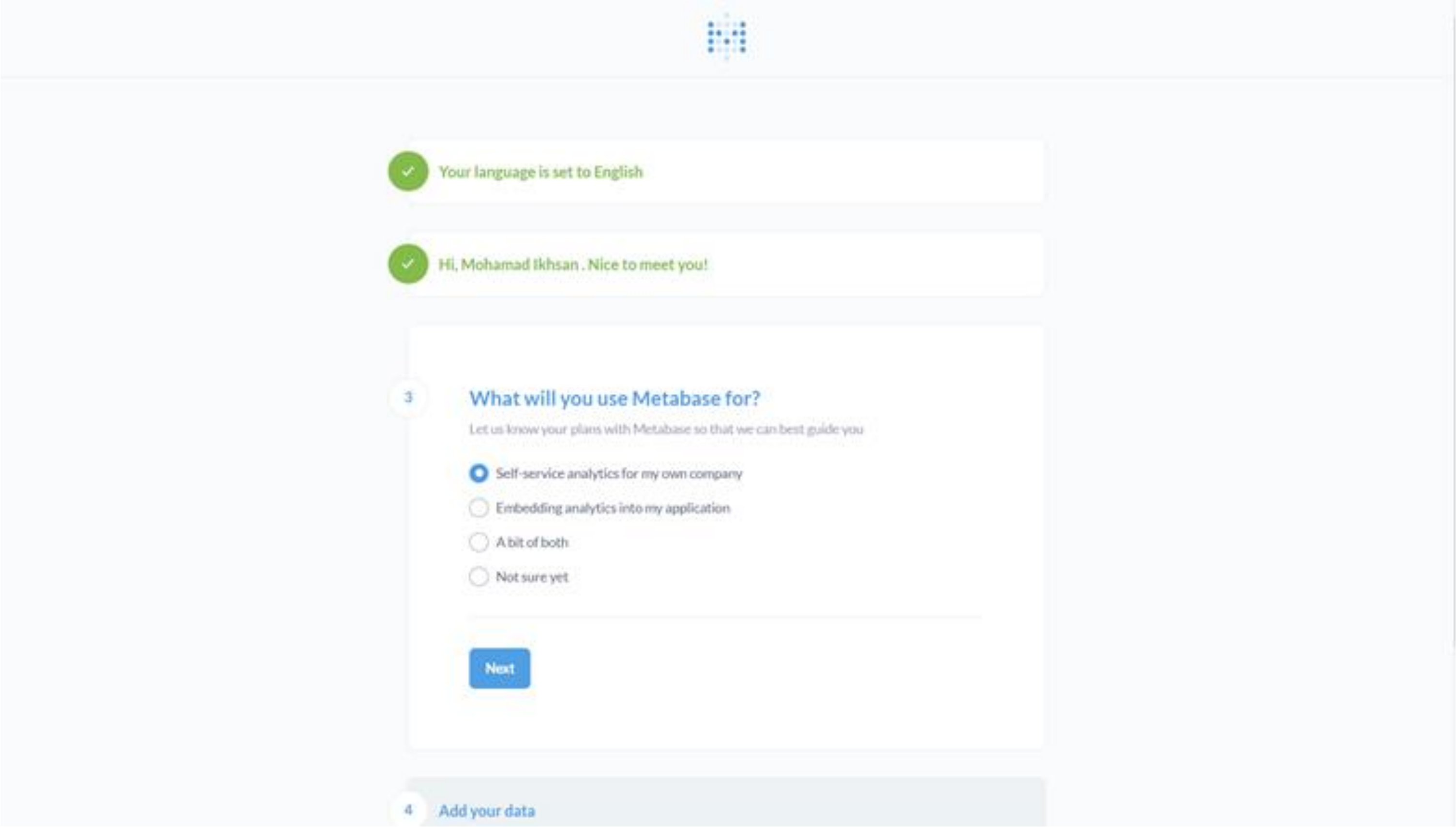
Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

User Information and Setting Up a Password

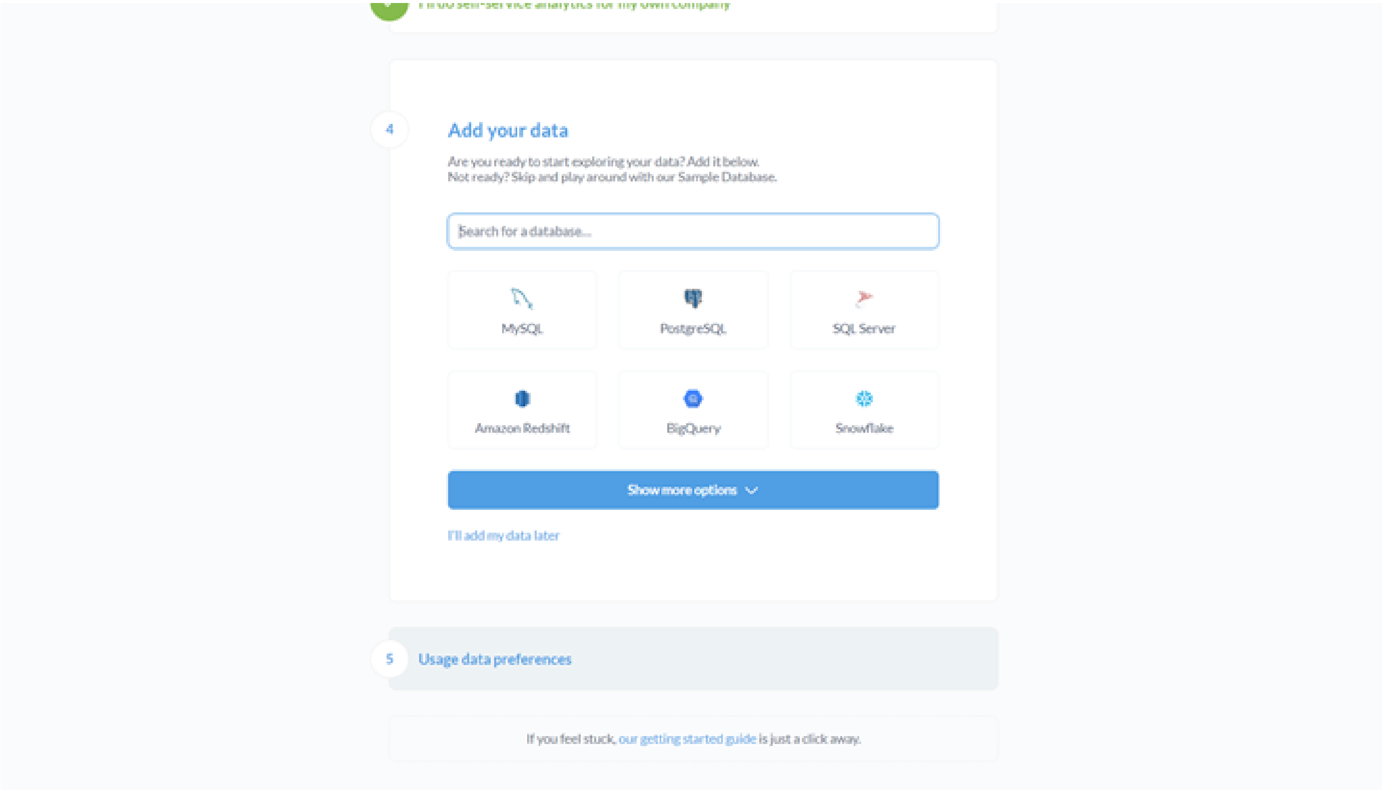


Initial setup steps for Metabase, including selecting the preferred language, entering user information, and setting up a password.

What will you use Metabase for?

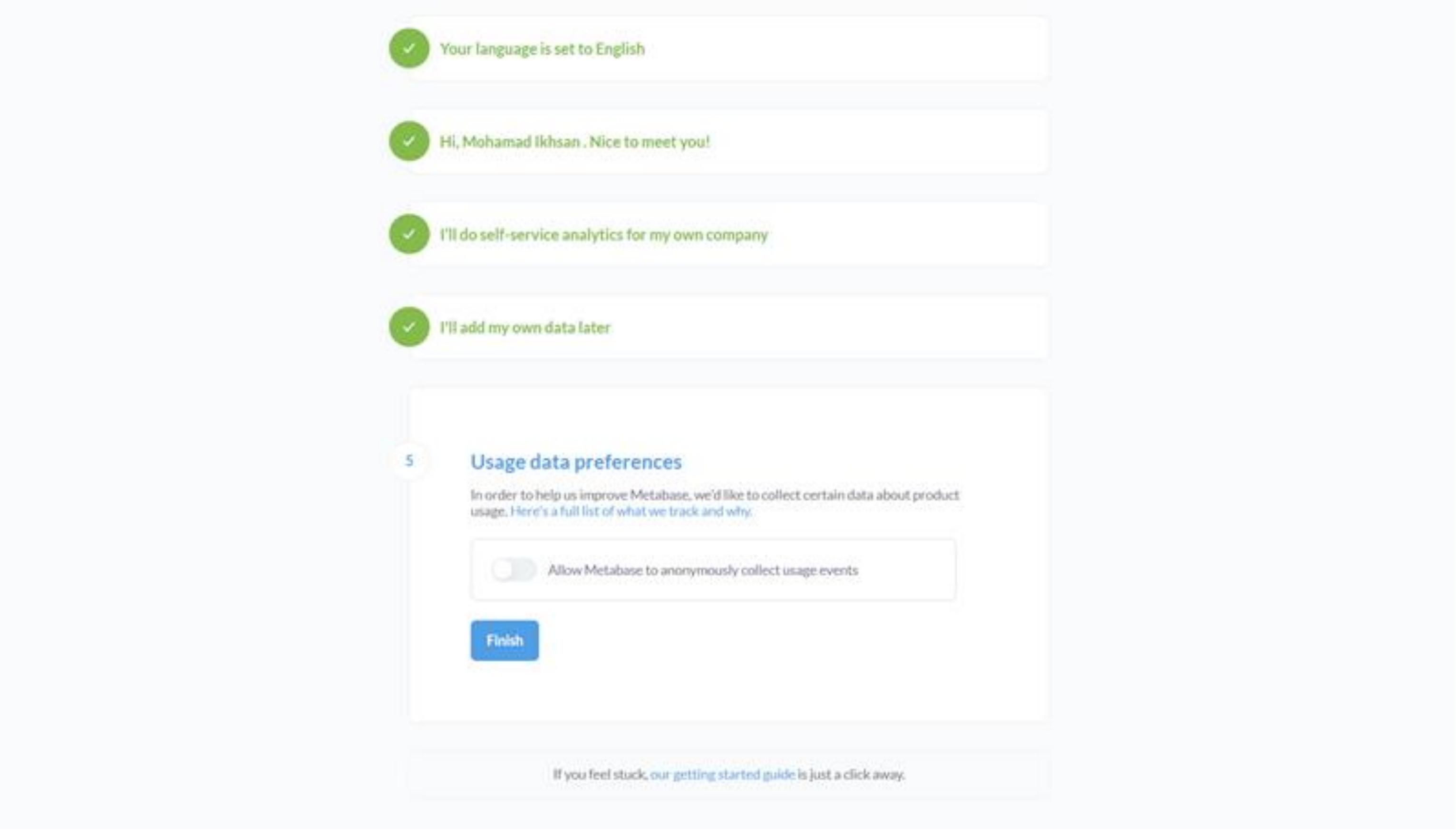


Add Data Connection

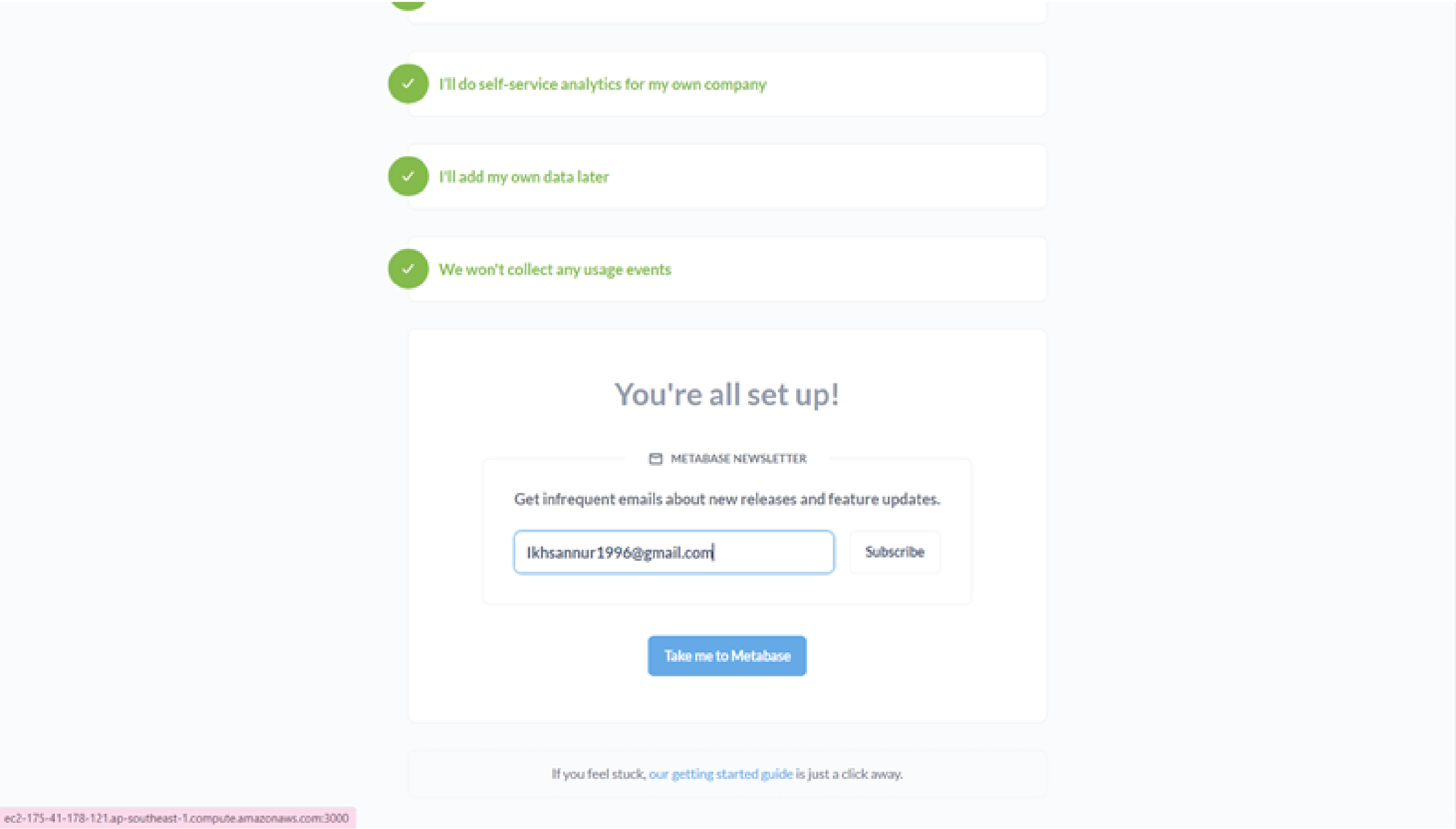


Options for adding data sources to Metabase, including popular databases like MySQL, PostgreSQL, SQL Server, Amazon Redshift, BigQuery, and Snowflake.

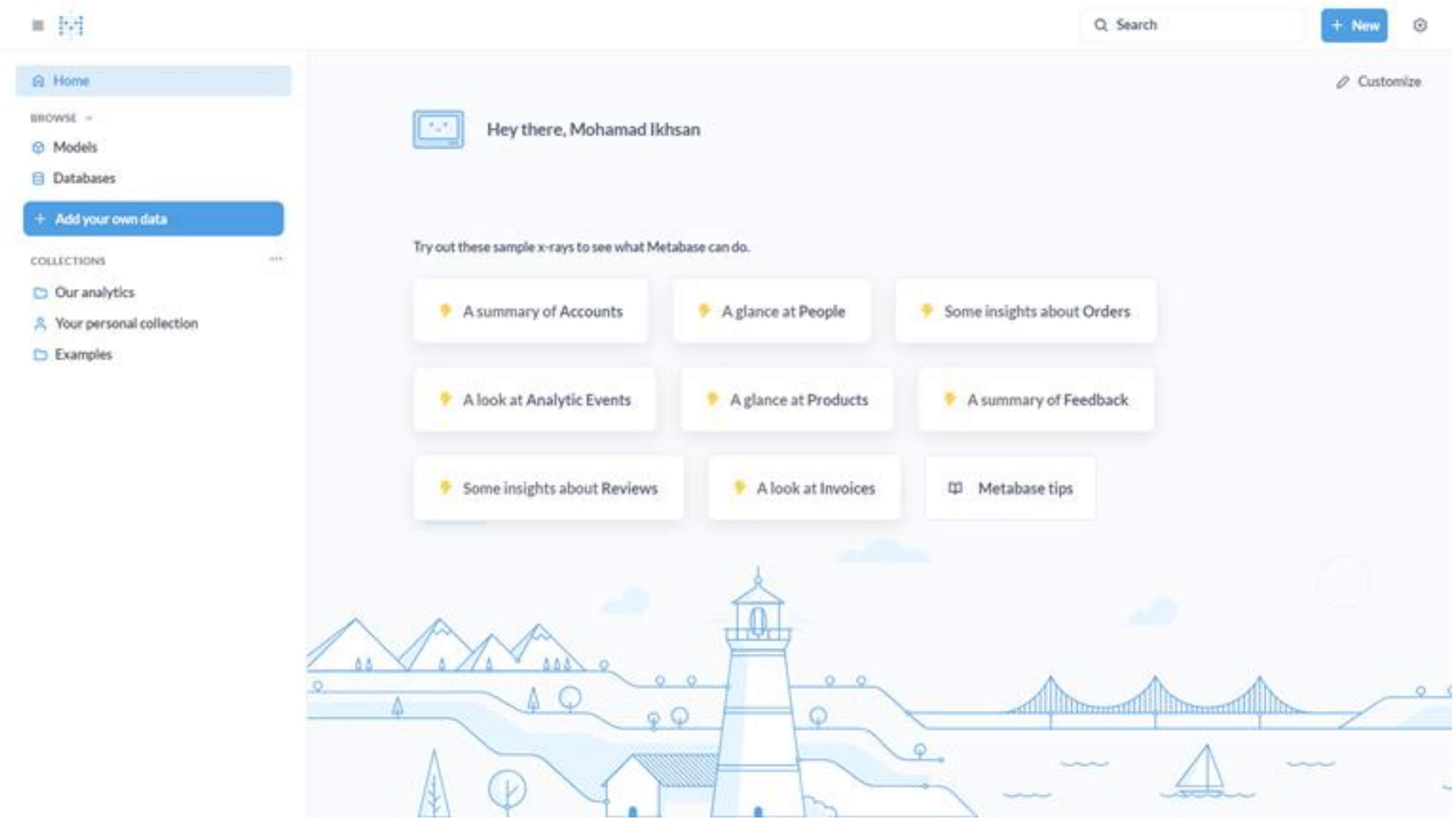
Usage Data Preferences



You’re All Set Up

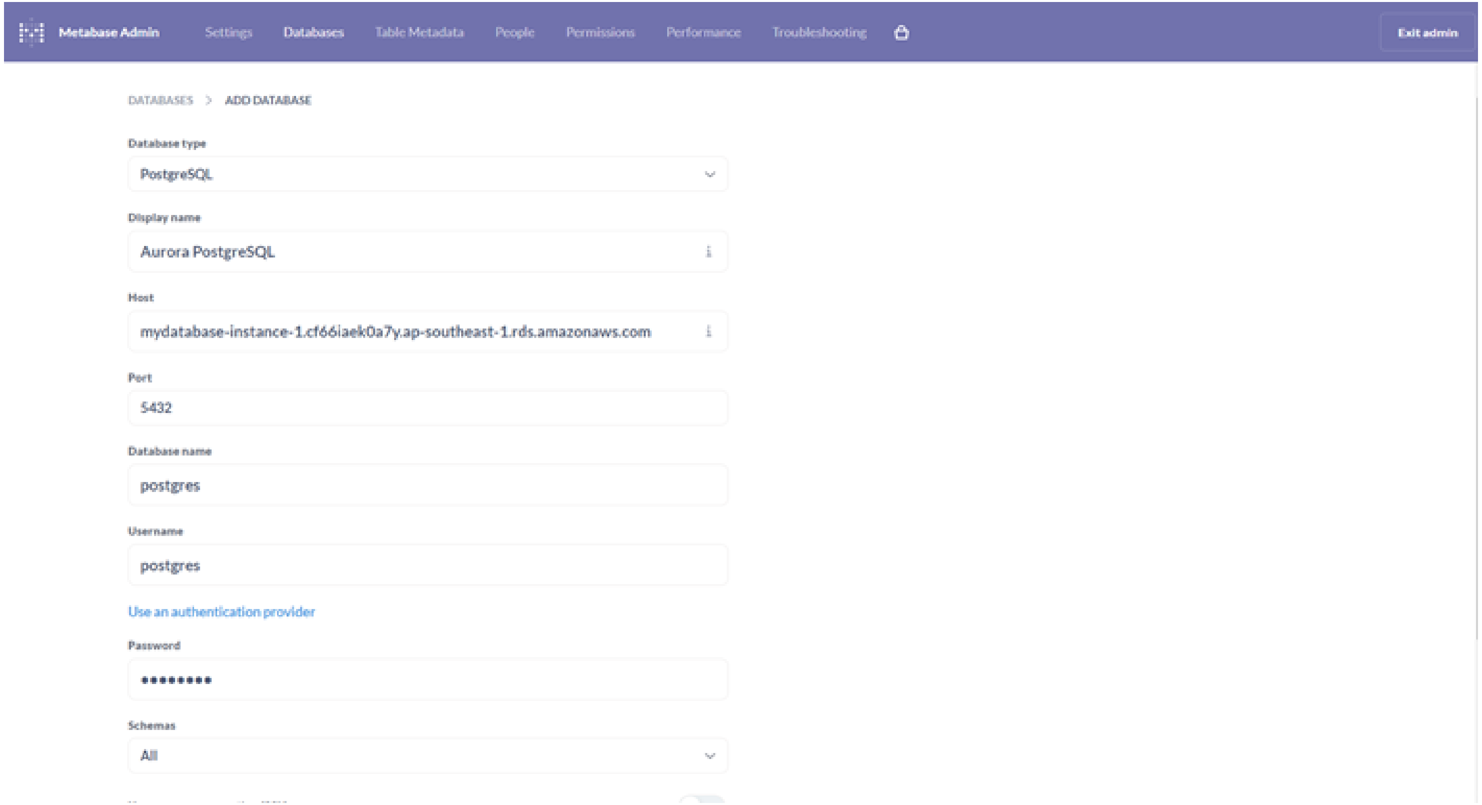


Metabase Home Page



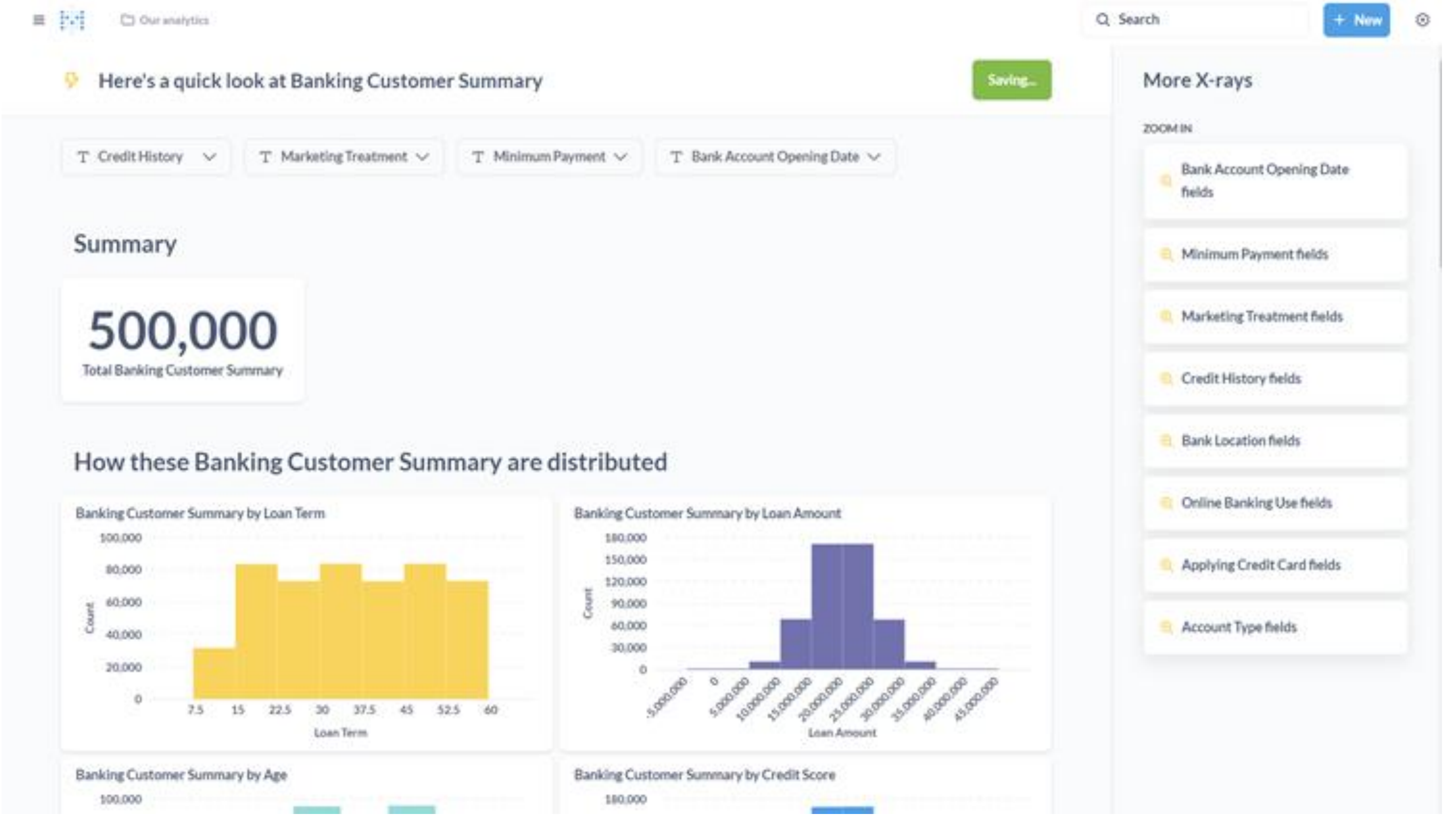
Overview of the Metabase dashboard, including sample data exploration options and a summary of various analytics insights.

Test Database Connection to Aurora PostgreSQL



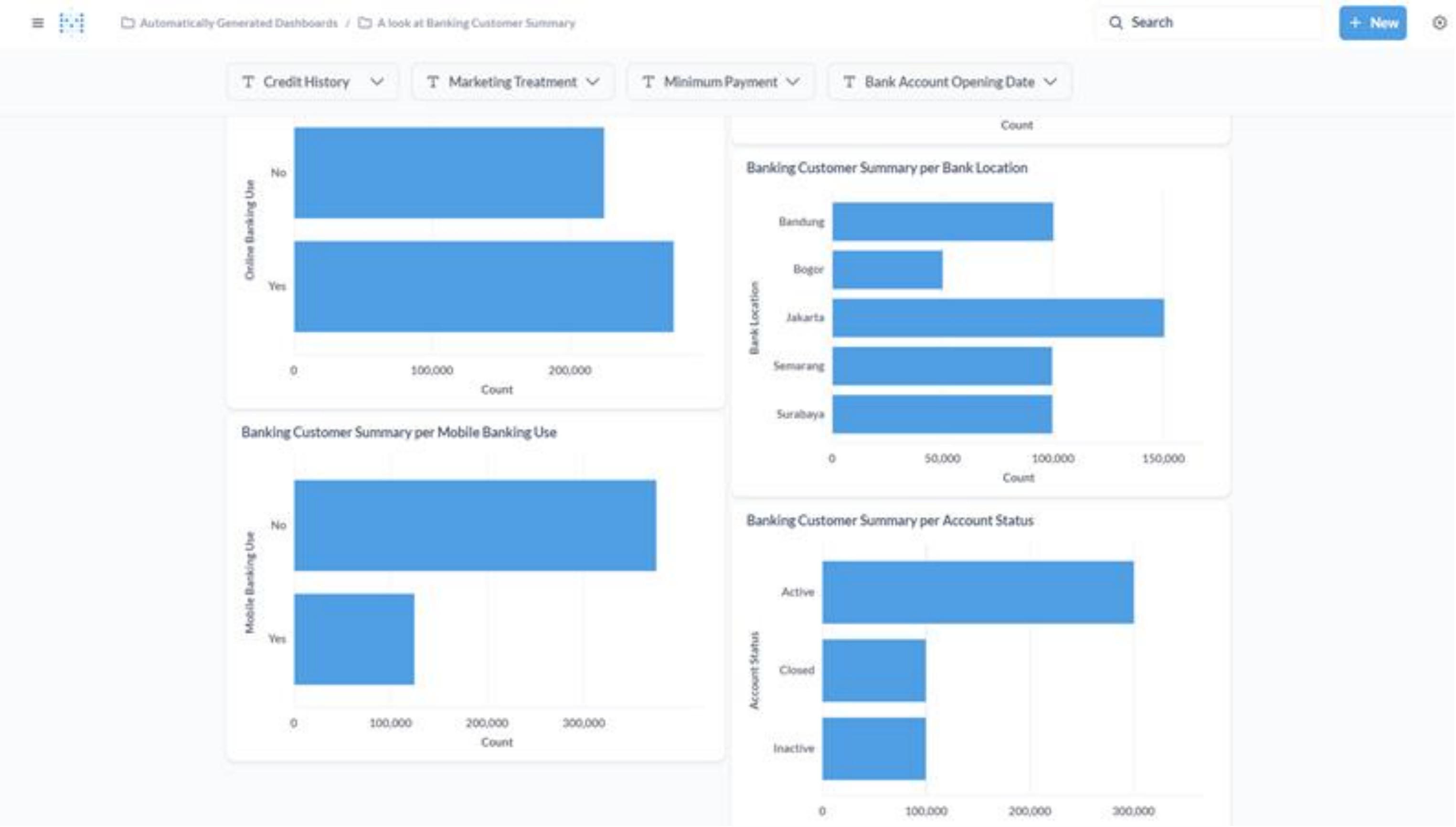
Steps to connect Metabase to a Aurora PostgreSQL database, including entering the host, port, database name, username, and password.

Metabase Dashboard Using X-Ray



Metabase has an X-ray feature that generates a summary or visualization of a table or column to provide users with a quick understanding of their data.

Metabase Dashboard Using X-Ray



A quick look at various analytics insights, such as banking customer summaries, loan terms, loan amounts, credit scores, and more.

Conclusion and Recommendation

Conclusion

- **Scalable Infrastructure:** Deploying Metabase on AWS EC2, combined with RDS Aurora PostgreSQL, provides a scalable and reliable data analytics platform.
- **Efficient Setup:** The step-by-step process ensures a smooth setup of both the database and Metabase, facilitating quick deployment.
- **Enhanced Data Visualization:** Users can leverage Metabase's capabilities for powerful data visualization and analytics.
- **AWS Integration Benefits:** The integration with AWS services ensures high availability, security, and performance for data-driven decision-making.

Recommendations

- **Security**: Use AWS Secrets Manager, enable database encryption, and manage access with security groups and IAM roles.
- **Backups and Monitoring**: Implement regular RDS backups with automated snapshots and set up CloudWatch for monitoring and alerts.
- **Cost Optimization**: Review and optimize AWS usage, selecting the right instance types and using reserved instances or savings plans.
- **Updates and Training**: Keep Metabase and AWS services updated, and ensure thorough documentation and team training.

Thank You